



Your Digital Innovation Partner

Technical Proposal

Higher
Colleges of
Technology



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العليا

Abu Dhabi, UAE

RFQ: NE-26-0129

Secure Access Service Edge (SASE)- 3 Years

Computer Network Systems-Sole Proprietorship L.L.C.

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02/09/2026

CNS Ref: FN-CNS/260870/OPP

Cover Letter

Attention to:

Higher Colleges of Technology- HCT

Subject: Secure Access Service Edge (SASE)- 3 Years

Dear Ma'am / Sir,

Computer Network Systems-Sole Proprietorship L.L.C., is pleased to present our proposal to Higher Colleges of Technology- HCT for their request for quotation on Secure Access Service Edge (SASE)- 3 Years

We thank you for inviting CNS to be part of your RFQ. Kindly find herewith our Technical Proposal.

We hope the enclosed proposal meets your expectation; and we look forward to our fruitful cooperation.

Should you require any additional information or have any queries, please do not hesitate to contact us.

Thanks and Regards,

Fadi Nasser

Senior Account Manager

Executive Summary

Computer Network Systems (CNS) is pleased to submit our proposal, in response to Higher Colleges of Technology- HCT Request for Secure Access Service Edge (SASE)- 3 Years. CNS has an excellent track record of successfully designing solutions, transitioning, managing and operating ICT, networking solutions, managed Services & engineering services. We will bring to Higher Colleges of Technology- HCT, our proven expertise, best practices, and deep industry experience that will capture and accelerate sustainable benefits.

We are excited about the opportunity and understand the strategic significance of this project to Higher Colleges of Technology- HCT's future success and we truly want to partner with you in this endeavor. While we believe that our proposal will best capture the benefits sought by Higher Colleges of Technology- HCT, we recognize that it will need to evolve as we build our relationship. Consequently, we would welcome to opportunity to further tailor our solution and relationship to meet your needs.

CNS spread across the region with offices in Abu Dhabi, Dubai, Oman, KSA and Kuwait providing a holistic approach that covers all aspects of information communication technology system integration, consultancy, operation, managed services, and IT outsourcing. CNS also has state-of-the-art network operation centers in Abu Dhabi and Dubai that develop in-house software and applications providing end-to-end solutions to our customers and partners.

CNS will use the best practices in the industry to fulfil Higher Colleges of Technology- HCT's requirement taking into consideration the availability & the Quality of Service, the total cost of ownership and the return on investment (ROI). CNS has the experience, the expertise and a long history of providing superior quality, value, and accountability. We have designed a practical customized solution that meets Higher Colleges of Technology- HCT needs.

CNS looks forward to working together with you and hopes that our solution and service offering is up to your requirements and expectations.

Our Commitment to Higher Colleges of Technology- HCT

CNS utilizes ITIL Best Practices and Processes, Subject Matter Expertise, Best-of-Breed Technologies, and an experienced, knowledgeable team to provide Higher Colleges of Technology- HCT ICT Operations Support. CNS' Best Practices and Processes represent our most innovative thinking compiled from our commercial and Government client engagements. CNS team is excited to present this proposal and looks forward to the opportunity to discuss it in detail with Higher Colleges of Technology- HCT

As you read the remainder of our proposal, we will expand on the following key factors that differentiate us from other vendors:

- The CNS team has a long and unparalleled history of successfully working and collaborating with our clients to deliver success stories. We always commit to staff the most experienced Subject Matter Experts and bilingual Consultants available in the marketplace on the project.
- We are confident in our team's ability to partner with Higher Colleges of Technology- HCT for the successful ICT Operations Support. Our team brings not just the right companies, but also the right people. We bring our finest technical team on-board. Our depth of knowledge, gained on several support services, will be essential to a smooth service and on cost.
- CNS has a long and sustained commitment implementing innovative ICT projects in the public sector for many years. We understand the challenges, the environment and associated risks. We reflect our confidence through our willingness to put our performance on the line.

TABLE OF CONTENTS

Description	Page Number
Cover Letter	2
Executive Summary	3
TABLE OF CONTENTS	4
Configuration Management	7
Statement of Confidentiality	7
1. Context & Executive Overview	8
2. Proposed Solution: Zscaler Internet Access (ZIA)	9
2.1. Understanding the ZIA Cloud Architecture	10
2.1.1. Zscaler Central Authority	10
2.1.2. ZIA Public Service Edges	10
2.1.3. Nanolog Clusters	11
2.2. Traffic Forwarding	11
2.2.1. Zscaler Client Connector	12
2.2.2. GRE Tunnels	12
2.2.3. IPSec Tunnels	12
2.2.4. PAC Files	13
2.3. Zscaler Zero Trust Firewall	13
2.3.1. Zscaler Cloud Native Firewall Key Capabilities	13
2.3.2. Zscaler Zero Trust Firewall Core Features:	14
2.3.3. Zscaler Zero Trust Firewall Licensing:	16
2.3.4. Cloud/Branch Connector	17
2.4. Zscaler Zero Trust Sandboxing	18
2.4.1. Zscaler Sandboxing Key Capabilities	18
2.4.2. Zscaler Sandboxing Core Features	20
2.4.3. Zscaler Sandboxing Licensing	21
2.5. Zscaler Zero Trust Browser (ZTBrowser)	22
2.5.1. Zscaler Cloud Browser Key Capabilities	22
2.5.2. How Zscaler Cloud Browser Works	22
2.5.3. Zscaler Zero Trust Enterprise Browser (EB)	23
2.6. Zscaler AI Guard for Users	23
2.6.1. Key Capabilities of AI Guard	24
2.6.2. AI Guard for Users Architecture	25
3. Proposed Solution: Zscaler Private Access (ZPA)	26
3.1. Key Capabilities	26
3.2. Getting Started with ZPA	26
3.3. ZPA Cloud Architecture	27
3.3.1. ZPA Central Authority	28
3.3.2. ZPA Public Service Edges and ZPA Private Service Edges	28
3.3.3. App Connectors	28
3.3.4. Logging Streaming Service	28

3.4.	Traffic Forwarding.....	29
3.4.1.	Zscaler Client Connector	29
3.4.2.	Browser Access	29
3.5.	Understanding Communication Between Components and Subsystems	30
3.5.1.	Zscaler Tunnel.....	30
3.5.2.	Microtunnel.....	30
3.6.	Transitioning to a Zero Trust ZPA Deployment	31
3.7.	AI-Powered Segmentation	32
3.8.	Zscaler VPN (For Legacy Apps).....	34
3.8.1.	ZPA Architecture for VoIP and Server-to-Client Apps: Network Connectors, Secondary Tunnel, and Zero Trust Exchange.....	34
4.	Proposed Solution: Zscaler Digital Experience Monitoring (ZDX)	35
4.1.	Key Capabilities	36
4.2.	Understanding ZDX Probes	36
4.2.1.	ZDX Web Probes	37
4.2.2.	ZDX Cloud Path Probes	38
4.3.	Applications Monitoring	39
4.3.1.	Applications Definition.....	40
4.3.2.	Defining Custom Applications and Probes for ZDX	40
4.4.	ZDX, ZPA and Private Applications	41
4.5.	Device Scoring and Device Events.....	41
5.	Bill of Quantity.....	43
5.1.	Identity Integration & ZCC Enrolment	43
5.2.	ZIA Business Edition Baseline	44
5.3.	Cyber Threat Protection — Sandbox & Firewall Advanced	44
5.4.	Secure AI Access	44
5.5.	Zero Trust Browser Advanced	44
5.6.	ZDX Advanced Plus.....	44
5.7.	Zscaler Private Access (ZPA)	44
6.	Assumptions	45
7.	Dependencies	45
8.	Out of Scope.....	45
9.	Place(s) of Performance	46
10.	Deliverables.....	46
11.	Miscellaneous.....	46
12.	Solution Implementation.....	47
12.1.	Implementation - DREAM Methodology.....	47
12.2.	Implementation & Migration Approach	48
13.	Project Management	49
13.1.	Project Structure	49
13.2.	Proposed Team Structure	49
13.3.	Project Plan — 3 Months	50

13.4.	CNS PMO History	50
13.5.	PMO Qualifications	51
13.6.	PMO Methodology.....	51
13.7.	PM Process	53
13.7.1.	Project Start-up.....	53
13.7.2.	Project Execution and Control	54
13.7.3.	Project Closure.....	55
14.	Service Delivery (Support and Maintenance).....	56
14.1.	What CNS Support Services are:	56
14.2.	CNS Customer Happiness Center	56
14.3.	CNS Resources.....	57
14.4.	Escalation Matrix	58
15.	Appendices.....	59
15.1.	Proposal Terms and Conditions	59
15.2.	CNS Company Profile	60
15.2.1.	Ghobash Group.....	61
15.2.2.	CNS Territory	61
15.2.3.	CNS Vision and Mission	62
15.2.4.	CNS Solution Key Differentiators	62
15.2.5.	CNS Value Proposition	63
15.2.6.	Our Recent Awards.....	63
15.2.7.	Our Portfolio.....	64
15.2.8.	Our Clients.....	64
15.2.9.	Partner and Alliances	65

Configuration Management

Configuration Management		
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Version Number	Issue Date	Changes
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Statement of Confidentiality

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1. Context & Executive Overview

Higher Colleges of Technology (HCT) is taking a visionary step in its digital transformation journey by seeking a unified, scalable, and resilient cloud-delivered Secure Access Service Edge (SASE) platform.

As the largest applied higher education institution in the United Arab Emirates, operating 17 campuses across Abu Dhabi, Dubai, and the Northern Emirates, HCT requires a modern cybersecurity posture capable of seamlessly protecting both managed staff devices and student Bring Your Own Device (BYOD) environments without sacrificing performance or operational agility.

Today's Threat Landscape presents unprecedented challenges: organizations face high-volume, highly sophisticated cyber threats, including ransomware, advanced phishing attacks, malware, and data exfiltration. These risks are exacerbated by cloud adoption, hybrid learning models, and direct-to-internet traffic flows. Legacy perimeter-based architectures and appliance-centric web gateways can no longer provide consistent security enforcement or holistic visibility for modern, distributed user populations.

To solve these challenges, **CNS** is delighted to present this proposal for **the Zscaler Zero Trust Exchange™** platform. Powered by **Zscaler Internet Access (ZIA)** and optionally augmented by **Zscaler Private Access (ZPA / ZTNA)**, our cloud-native solution eliminates the complexity of fragmented legacy security stacks, providing unified, AI-powered protection for every HCT campus, user, and device.

2. Proposed Solution: Zscaler Internet Access (ZIA)

Securing today's cloud- and mobile-first enterprise requires a fundamentally different approach built on zero trust. Zscaler Internet Access, part of the Zscaler Zero Trust Exchange™, is the world's most deployed security service edge (SSE) platform, built on a decade of secure web gateway leadership.

Delivered as a scalable and resilient SaaS cloud security platform, ZIA eliminates legacy network security solutions to stop advanced attacks and prevent data loss with a comprehensive zero trust approach, offering:

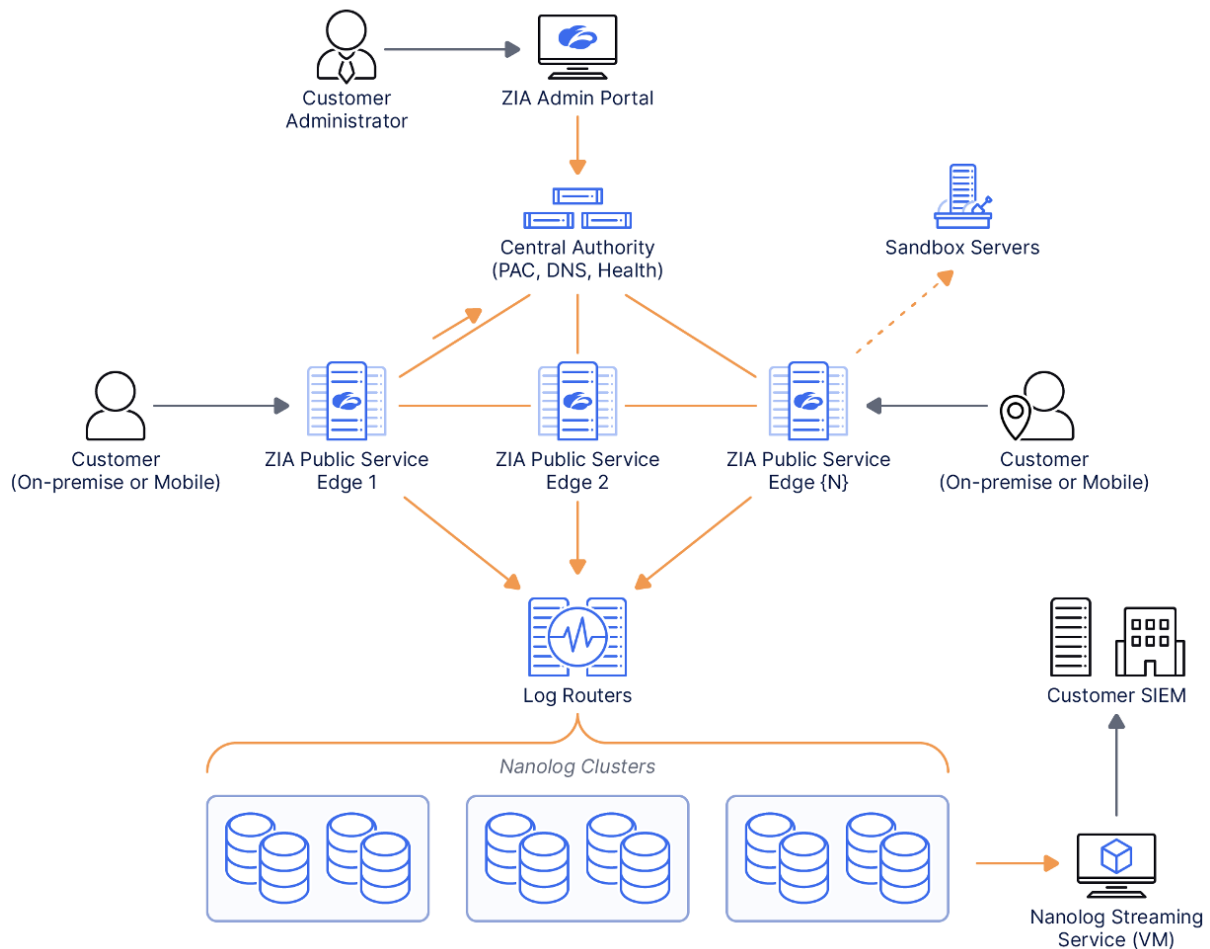
- **Best-in-class, consistent security for today's hybrid workforce:** When you move security to the cloud, all users, apps, devices, and locations get always-on threat protection based on identity and context. Your security policy goes everywhere your users go.
- **Lightning-fast access with zero infrastructure:** Direct-to-cloud architecture ensures a fast, seamless user experience. This eliminates backhauling, improves performance and user experience, and simplifies network administration—with no physical infrastructure required.
- **AI-powered protection from the world's largest security cloud:** Inline inspection of all internet and SaaS traffic, including SSL decryption, with a suite of AI-powered cloud security services to stop ransomware, phishing, zero-day malware, and advanced attacks based on threat intelligence from 500 trillion daily signals.
- **Simplified management:** Using a cloud native security solution infused with AI, less hardware to manage, automation to streamline workflows, and business-focused policy creation frees up valuable time for your team to focus on strategic goals.

Zscaler Internet Access includes a comprehensive suite of AI-powered security and data protection services to help you stop cyberattacks and data loss. As a fully cloud-delivered SaaS solution, you can add new capabilities without any additional hardware or lengthy deployment cycles. The modules available as part of Zscaler Internet Access are:

- **Cloud Secure Web Gateway (SWG):** Deliver a safe, fast web experience that eliminates ransomware, malware, and other advanced attacks with real-time, AI-powered analysis and URL filtering.
- **Cloud Access Security Broker (CASB):** Secure cloud apps with integrated CASB to protect data, stop threats, and ensure compliance across your SaaS and IaaS environments.
- **Cloud Data Loss Prevention (DLP):** Protect data in motion with full inline inspection and advanced measures like exact data match (EDM), optical character recognition (OCR), and machine learning.
- **Zscaler Firewall & cloud IPS:** Extend industry-leading protection to all ports and protocols and replace edge and branch firewalls with a cloud native platform.
- **Zscaler Sandbox:** Stop never-before-seen and elusive malware across web and file transfer protocols with AI-driven quarantine, sharing consistent and global protection across all users in real time.
- **AI-Powered Zero Trust Browser:** Make web-based attacks obsolete and prevent data loss by creating a virtual air gap between users, the web, and SaaS.
- **Zero Trust Branch Connectivity:** Reduce risk and complexity with non-routable branch and data center connectivity for users, servers, and IOT/OT devices.
- **DNS Security:** Optimize DNS security and performance for all users, devices, and applications, on all ports and protocols, anywhere in the world.

2.1. Understanding the ZIA Cloud Architecture

Zscaler operates the world's largest security-as-a-service (SECaaS) cloud platform to provide the industry's only 100% cloud-delivered web and mobile security solution. The highly scalable, global, multi-cloud infrastructure features three key components: the Zscaler Central Authority (CA), ZIA Public Service Edges, and Nanolog clusters.



2.1.1. Zscaler Central Authority

The Zscaler Internet Access (ZIA) Central Authority (CA) is the brain and nervous system of a Zscaler cloud. It monitors the cloud and provides a central location for software and database updates, policy and configuration settings, and threat intelligence. The CA consists of one active server and two servers in passive standby mode. The active CA replicates data in real time to the two standby CAs, so any of them can become active at any time. Each server is hosted in a separate location to ensure fault tolerance.

2.1.2. ZIA Public Service Edges

ZIA Public Service Edges are full-featured, inline internet security gateways that inspect all internet traffic bi-directionally for malware and enforce security and compliance policies.

An organization can forward its traffic to any ZIA Public Service Edge in the world or use the advanced geo-IP resolution capability of Zscaler to direct its users' traffic to the nearest ZIA Public Service Edge.

When the user moves to a different location, the policy follows the user, with the ZIA Public Service Edge downloading the appropriate policy. Customer traffic is not passed to any other component within the Zscaler

infrastructure. The TCP stack on the ZIA Public Service Edge runs in user mode and is specially crafted to ensure multitenancy and data security. ZIA Public Service Edges never store any data to disk.

Log data generated for every transaction is compressed, tokenized, and exported over secure TLS connections to Log Routers that direct the logs to the Nanolog cluster, hosted in the appropriate geographical region, for each organization.

ZIA Public Service Edges are always deployed in active-active load balancing mode all over the world, and the CA monitors the health of ZIA Public Service Edges to ensure availability.

2.1.3. Nanolog Clusters

Nanolog clusters store transaction logs and provide reports. Each cluster consists of one active server and two servers in passive standby mode.

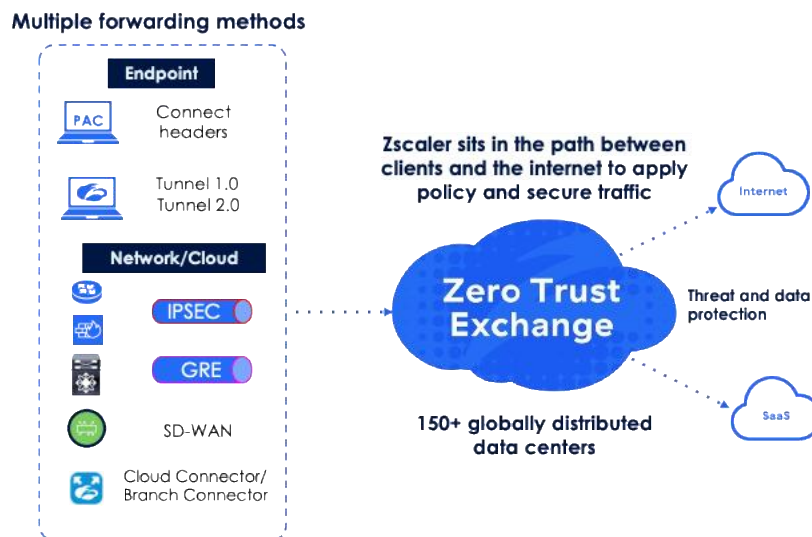
The active Nanolog immediately replicates data to the other two servers, so any of them can become active at any time, with no data loss. Each Nanolog server is hosted in a separate location to ensure fault tolerance.

Every second, a Nanolog cluster receives logs from all over the world, correlates them to a specific customer organization, and writes them to disk for high-speed retrieval of reporting and analytics.

A Nanolog cluster processes up to 1.2+ billion logs per day. Additionally, Zscaler offers a Nanolog Streaming Service (NSS), which uses a virtual appliance to stream web and firewall traffic logs in real time from the Zscaler Nanolog to the customer's security information and event management (SIEM) system.

2.2. Traffic Forwarding

Zscaler supports the following traffic forwarding mechanisms.



Depending on your environment and requirements, you can choose one or a combination of the following traffic forwarding methods. Zscaler recommends that you use a combination of tunneling, PAC files, Zscaler Client Connector, and Zscaler Cloud Connector, to forward traffic to the Zscaler service.

2.2.1. Zscaler Client Connector

Zscaler recommends that you install Zscaler Client Connector on your users' devices to protect their web traffic when they are accessing the internet from an unknown location (i.e., remote user).

With Zscaler Client Connector's internet security feature, you can protect your users' web traffic even when they are outside your corporate network. The app forwards user traffic to the Zscaler service and ensures that your organization's security and access policies are enforced wherever they might be accessing the internet from. The app automatically detects when the user is no longer within the secure network, then automatically creates a connection to the Zscaler service from an unknown location.

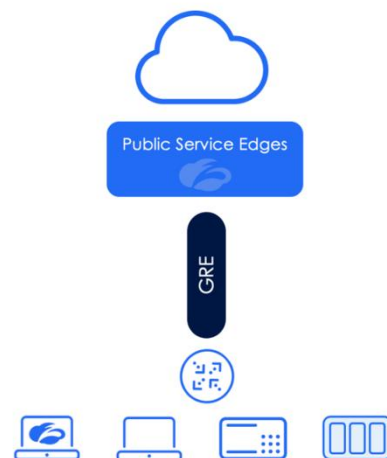


2.2.2. GRE Tunnels

A GRE tunnel can be used for forwarding internet-bound traffic from your corporate network to the Zscaler service.

If your organization has an internal router or switch that supports GRE and its egress port has a static address, you can configure a GRE tunnel.

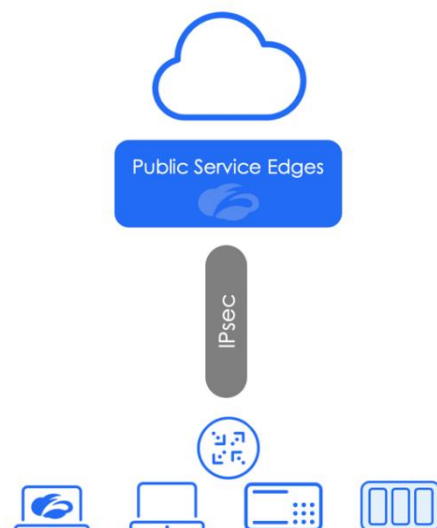
Zscaler recommends that you configure two GRE tunnels from an internal router behind the firewall to the ZIA Public Service Edges: a primary tunnel to a ZIA Public Service Edge in one data center and a secondary tunnel to a ZIA Public Service Edge in another data center. These deployments provide visibility into internal IP addresses, which can be used for security policies and logging.



2.2.3. IPSec Tunnels

IPSec is a suite of protocols that provide network-layer security to a Virtual Private Network (VPN). If your router does not support GRE or if you use dynamic IP addresses, you can configure IPSec VPN tunnels.

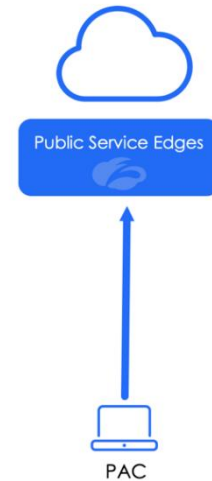
Zscaler recommends that you configure two IPSec tunnels from an internal router behind the firewall to the ZIA Public Service Edges: a primary tunnel to a ZIA Public Service Edge in one data center and a secondary tunnel to a ZIA Public Service Edge in another data center. These deployments provide visibility into internal IP addresses, which can be used for security policies and logging.



2.2.4. PAC Files

A PAC file is a text file that directs the browser to forward traffic to a ZIA Public Service Edge. You can install a PAC file for each of your users to ensure coverage outside the corporate network.

If certain applications need to be excluded from the Zscaler service, it is easy to write an exclude statement in your PAC file to send traffic directly. Using PAC files allows you to set exceptions for forwarding your traffic, such as excluding intranet corporate sites from the Zscaler service.



2.3. Zscaler Zero Trust Firewall

Zscaler Zero Trust Firewall delivers cloud-based protection for non-web traffic (DNS, FTP, RDP, SSH, Telnet and more), while Zscaler Internet Access (ZIA) delivers deep inspection for web traffic (HTTP/HTTPS). Together, they secure all ports and protocols including non-web applications and non-standard web, across all users, devices and locations.

Zero Trust Firewall improves connectivity and performance by enabling local internet breakout—eliminating the need for VPN backhauling or redundant on premises appliance stacks.

Zero Trust Firewall helps organizations easily meet regulatory standards while universally configuring, managing, and enforcing user- and application-aware threat protection and risk-based policies to ensure network and application visibility with a centralized policy management console. As a true firewall-as-a-service (FWaaS) solution—built from the ground up for the cloud—Zscaler Zero Trust Firewall eliminates the need for hardware provisioning or managed service models, delivering elastic scale and policy control without legacy limitations.

Zscaler Zero Trust Firewall provides forensically complete logs for every session, capturing detailed attributes such as user identity, device info, source and destination IPs, port, country, threat category, and more. This rich visibility across all users, devices, and locations eliminates the need for packet capture tools and ensures organizations have access to the right information at the right time.

2.3.1. Zscaler Cloud Native Firewall Key Capabilities

- **Purpose-Built For Today's Digital World:** Zscaler Zero Trust Firewall ensures you can securely access the internet and handle all web and non-web traffic, across all ports and protocols, with infinite elastic scalability and unbeatable performance. Your users get consistent protection no matter what device they're using or where they are—at home, HQ or branch offices, or on the road—without the cost, complexity, and performance limitations of traditional network security and next-generation firewall appliances.
- **Powered By an Adaptive Zero Trust Platform:** Stop compromising for static inspections, performance degradation, and capacity limits from physical firewall appliances. Built on a fully integrated, cloud-native platform, Zscaler Zero Trust Firewall elastically scales to handle cloud application traffic requiring long-lived connections while natively intercepting and inspecting TLS/SSL traffic—at scale—to detect malware hidden in encrypted traffic.
- **Transformative Hybrid and Branch Connections:** Evolve from costly and network-centric infrastructure to true cloud-delivered local internet breakouts. Route internet traffic locally to provide direct-to-cloud

connections for consistently fast connections while delivering security and access controls for all ports and protocols. Without the need for any appliances to deploy or manage, this reduces MPLS backhauling costs and eliminates expensive and time-consuming patch management, coordination of outage windows, and policy management.

- **Ubiquitous Security for Modern Workforces:** Leverage real-time security updates informed by 500+ trillion daily signals and shared across the entire cloud each day for identical protection on any device wherever users connect. Bringing the entire security stack close to the user provides unparalleled user- and app-aware threat protection with dynamic, follow-me policies on and off the corporate network.
- **Always-On Blocking of Known Malicious Attacks:** Go where traditional solutions can't with a cloud-delivered intrusion prevention system (IPS) managed by Zscaler ThreatLabz. Through unlimited inline traffic inspection, including IoT/OT and encrypted traffic, behavioral IPS signatures are applied in real time across thousands of web-based and non-web applications.
- **DNS Optimization for Performance and Security:** Achieve faster resolution by pairing geographically local apps, driving better user experience and cloud app performance while implementing Domain Name System (DNS) security and control policies. With TLS/SSL inspection at scale, gain back visibility and stop attackers from abusing DNS-over-HTTPS (DoH), better protecting users and employees from reaching malicious domains and bypassing enterprise policies. By delivering DNS-as-a-service, Zscaler minimizes latency and secures local internet breakouts using full proxies for all DNS traffic and leverages machine learning to detect and block data exfiltration tunnel activity.
- **Easy-To-Understand Policy Management:** Universally define, deploy, and immediately enforce policies for all users, across all locations from a single console. In place of complex matrices of policy, network configurations and recreating policies for each location of typical firewalls, Zero Trust Firewall simplifies policy management by centralizing granular firewall rules based upon user, application, location, group, and department.
- **Enforce Precise Policies with Process-Level Awareness:** Extend Firewall, DNS, and IPS controls beyond IPs and ports by identifying the actual application process running on the device - such as PowerShell.exe or Chrome.exe. This enables security teams to block or allow activity based on the true source of the connection, reducing false positives, enhancing threat detection, and allowing more targeted risk response.

2.3.2. Zscaler Zero Trust Firewall Core Features:

Feature Name	Description
Centralized policy management	Define and immediately enforce policies across all locations without the need to recreate policies for each location
Fully integrated security services	Share contextual information across DLP, APT, sandbox, and other services to provide better protection and deeper visibility
Real-time granular control, logging, and visibility	Maintain forensically rich logging for detailed visibility with globally unified and unlimited logging for six months, enabling analysis and correlation for trend discovery, productivity analysis, and troubleshooting
User-aware threat protection	Define users by Groups, Departments, or Locations, including setting work-from-home or remote users as a location, and integrate with identity providers and local user databases, allowing consistent policies regardless of users' physical locations
Endpoint App Control	Define and enforce policies based on the actual application process running on the device, such as PowerShell or Chrome, enabling more precise control, reducing false positives, and providing stronger protection against targeted threats and insider activity
	Supports a wide range of application types:

App-aware threat protection	- All network services - ports and protocols - Network applications identified by SNI (hostname) and Deep Packet Inspection (DPI) - Application Services such as UCaaS, using First Packet Identification, IP address, FQDN groups, and other heuristic-based detection methods
Advanced security inspection	Apply advanced deep-packet inspection on non-web protocols, including FTP, DNS, RDP, Telnet, and more to identify and prevent evasive traffic on non-standard ports
Adaptive IPS security with custom rules	Deliver always-on threat prevention with a cloud-delivered IPS engine powered by thousands of adaptive, behavioral, and context-aware signatures managed by Zscaler ThreatLabz. Gain additional control with Cloud Custom IPS, allowing you to define and enforce Snort-based rules tailored to specific applications, threats or compliance needs.
End user notifications (EUN)	Communicate security policy actions to end users through desktop pop-ups via Client Connector for Firewall, DNS, and IPS, or browser-based messages for DNS blocks with web EUN, helping users understand why access was denied and reducing IT support queries
Granular role-based access control (RBAC)	Create and assign custom admin roles—such as SecOps or NetOps—with controls over who can view, edit, or manage security policies, logs, and alerts to enhance operational clarity and enforce least-privileged access across teams
DNS security and control	Optimize cloud application performance and minimize latency while ensuring uncompromised security by proxying all DNS through Zscaler. Enable policies based on user, app, location, and resolved IP country to automatically block users from malicious domains and detect and prevent DNS tunneling Resolution: DNS-as-a-service provides optimal resolution with localization, tenancy, and lowest latency DNS Filtering: Create custom DNS filtering rules to block, allow, or redirect different types of DNS requests against known and malicious destinations Security and Data Exfiltration: Detect malware, phishing, DNS tunneling, and data exfiltration using AI DNS over HTTPS (DoH): Prevent DoH blind spots and bypassing of organizational controls when encrypting DNS connections in common HTTPS traffic
Fully qualified domain name (FQDN) policies	Easily configure and manage access policies for applications hosted across multiple IPs
File Transfer Protocol (FTP) control and Network Address Translation (NAT) support	Support access control of FTP and FTP over HTTP and support for NAT destination proxy and NAT forwarding

Globally shared protection	Leverage the cloud effect: every time a new threat is identified in any of the tens of billions of requests processed daily by the Zscaler cloud, it gets blocked for all Zscaler users, everywhere
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2.3.3. Zscaler Zero Trust Firewall Licensing:

Zscaler provides two levels of license for our firewall, standard, and advanced.

Zscaler is proposing the advanced firewall license for Higher Colleges of Technology (HCT) to cover advanced firewalling requirements as well as DNS security requirements.

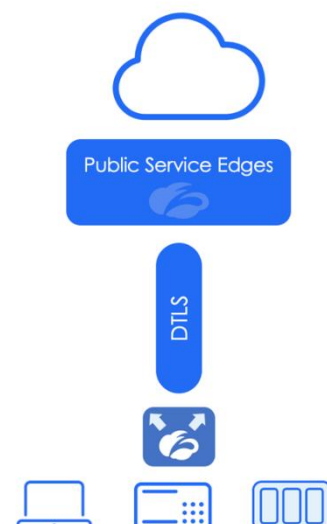
	Standard	Advanced
Zero Trust Firewall Policy Criteria		
Network and Application Services	 Up to 10 rules	
Source/Dest L3/4 and FQDN-based policies		
Location Awareness		
User Awareness	-	
Network Application (DPI)	-	
Dynamic Risk-Based Policy	-	
Geo-blocking (source & destination country)		
Firewall Rules	 Up to 10 rules	 Up to 10,000+ rules
Granular Role-Based Access Control (RBAC)		
End User Notifications (EUN)	 Web notifications for any DNS action	 ZCC notifications for any Firewall, DNS and IPS action
Endpoint App Control for process precision	-	

DNS Control		
Trusted DNS Resolver	✓	✓
DNS Filtering and Security	✓ Up to 64 rules	✓
DNS Tunnel and App Detection	-	✓
Ips Control with Custom Rules	-	✓
Ftp Control	✓	✓
Nat Control	✓	✓
Platform Features		
Full SSL Inspection	✓	✓
Real-time Logging	✓ Aggregated logging details for allow firewall actions and detailed logging details for block actions with full DNS logs	✓ All logs for all actions and all functions including user ID, App ID, IPS, and more

2.3.4. Cloud/Branch Connector

Zscaler Cloud/Branch Connector works similarly to Zscaler Client Connector, using software-defined networking (SDN) principles. It relies on a unified Zero Trust Architecture for connectivity, security, and access to the Zscaler cloud. It works by establishing zero trust access for server-to-server traffic, east-west traffic in a public cloud, IT/OT networks, and data centers.

Cloud Connector simplifies traffic forwarding to the Zscaler service by securing server access to the internet and providing highly secure and simplified access to the Zscaler cloud. Using Cloud Connector, you can forward specific web traffic or application traffic through the Zscaler cloud.



2.4. Zscaler Zero Trust Sandboxing

Built on a cloud native, proxy-based architecture, Zscaler Cloud Sandbox is the world's first AI-driven inline malware prevention engine that blocks unknown, file-based threats before they reach endpoints. With real-time analysis, instant AI-driven verdicts, and continuous

threat intelligence sharing, our sandbox helps organizations stay ahead of sophisticated attacks without compromising productivity.

To cater to varying customer needs, the Zscaler Cloud Sandbox offers two options:

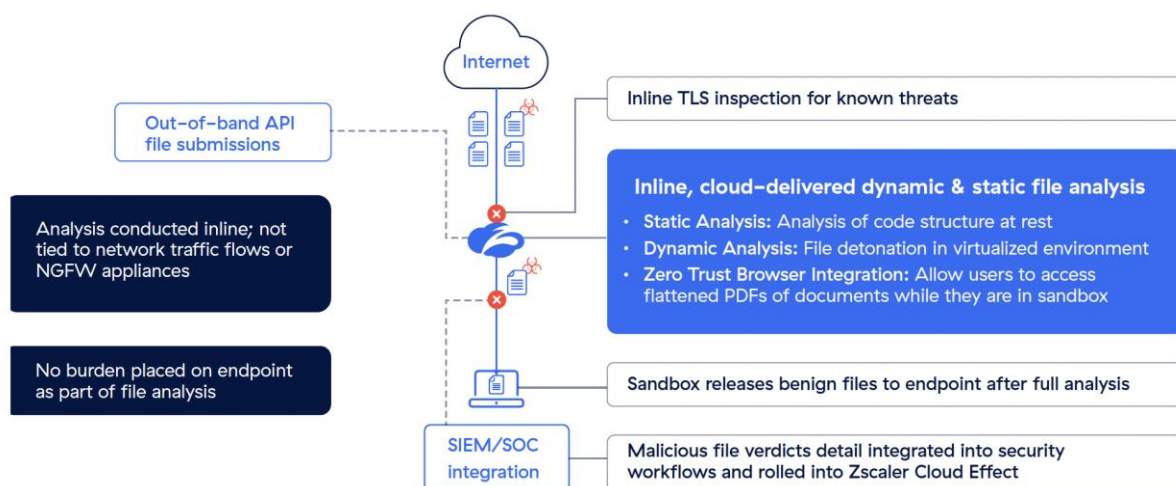
- Cloud Sandbox included with Zscaler Internet Access (ZIA), provides essential threat prevention capabilities for baseline protection.
- Advanced Cloud Sandbox delivers enhanced features such as AI Instant Verdict, granular policy control, API-driven analysis, and in-depth threat investigation for customers requiring more robust defenses.

Unlike traditional sandbox solutions that depend on network firewalls or endpoint-heavy agents, Zscaler Cloud Sandbox operates directly from the cloud. It eliminates the burden on endpoints, requires no reliance on network traffic flows, and seamlessly integrates into SOC workflows for faster, more effective threat detection and response.

The sandbox operates in two key modes to ensure flexible and comprehensive security:

- **Inline Inspection:** Files are analyzed as traffic flows through the cloud, and AI delivers real-time verdicts to block threats without latency.
- **Out-of-Band Inspection:** Files can be submitted via APIs for a quick scan or in-depth analysis as needed.

By offering foundational and advanced options, Zscaler ensures that customers of all sizes can benefit from layered security tailored to their needs.



2.4.1. Zscaler Sandboxing Key Capabilities

- **AI-Powered Security:** Harness the power of AI/ML models trained on over 600 million samples to deliver instant, high-confidence verdicts on unknown files. The AI Instant Verdict engine predicts whether a file is malicious or benign in real time, allowing immediate blocking or rapid delivery to users without delay. For cases where files are flagged as "suspicious," our customizable Minimum Threat Score enables users to fine-tune policy behavior, balancing security, and productivity. All confirmed threats are further validated through the Advanced Cloud Sandbox, which generates detailed reports and comprehensively analyzes the threat.
- **Layered Malware Detection:** Zscaler Cloud Sandbox delivers multilayered threat detection powered by real-time updates from over 500 trillion daily threat signals and the Zscaler Cloud Effect, which instantly

blocks known threats for all customers once identified. The process begins with static analysis to inspect the file's code structure while at rest, followed by dynamic analysis where the file is detonated in a virtualized environment to observe its behavior. A secondary static analysis examines any hidden or secondary payloads.

This layered approach, combined with inline protection and unlimited TLS/SSL inspection using our Single Scan, Multi-Action engine, ensures low-latency, high-performance threat prevention for even the most sophisticated, evolving threats



- Zero Trust Browser Integration:** The Zero Trust Browser integrates seamlessly with the Advanced Cloud Sandbox to ensure users remain productive without compromising security. This feature allows users to securely interact with original files in a cloud-based, isolated environment while the files undergo sandbox analysis. Risky or unknown files are automatically quarantined, but users can still access them securely during the analysis. If the sandbox analysis determines the file is safe, users can download the original version without delays.

If the file is flagged as malicious, users have options: they can download a flattened PDF version with all active content removed (CDR- 1) or, when appropriate, download an editable version with only the harmful content removed (CDR-3, may require additional licensing). This flexible integration ensures that users remain productive while zero-day threats are thoroughly examined and blocked when necessary. It is important to note that Zero Trust Browser is an optional module and works in tandem with the Advanced Cloud Sandbox to enhance the overall user experience.

- API-Driven Analysis:** Zscaler Cloud Sandbox's API integration streamlines investigations by allowing files to be sent directly to the sandbox for analysis. Security teams can easily access sandbox results via APIs, ingest them into SIEMs, SOAR platforms, or share with EDR solutions, enabling seamless SOC workflows. After detonation, the sandbox generates comprehensive reports, highlighting polymorphism, callback behaviors, and attack lifecycles, mapped to the MITRE ATT&CK framework. These insights can be operationalized to enhance threat detection, response, and proactive threat hunting.

For flexibility, organizations can use the Advanced Sandbox Submission API for thorough analysis of files out-of-band supporting on demand submission and reporting of unknown threats, or the Deep Inspection Scan API to receive an instant verdict on a file sample. Both APIs help reduce investigation times and enhance collaboration across the security stack.

- Granular Policy Control and Reporting:** With Zscaler Cloud Sandbox, administrators can configure and enforce granular security policies tailored to specific user roles, locations, or application categories— all within just a few clicks. The intuitive interface allows for fast, simple policy creation and deployment, minimizing time and resource investment. Policies, including rule orders for precise execution, are easily managed and automatically follow users or groups, even when they change locations. For deeper control, admins can configure custom hash blocklists, YARA rules, and automated JA3 fingerprinting to enhance threat detection.

2.4.2. Zscaler Sandboxing Core Features

Capabilities	Description
Prefiltering analysis engine	AV, file hash blocklists, YARA rules, automated JA3 fingerprinting detections, and ML/AI model
Static, dynamic, and secondary static analysis	All files go through 3 stages of sandbox analysis: static analysis, dynamic analysis, and secondary static analysis of all files dropped during dynamic analysis
File support	Check the following documentation for the list of supported file types: https://help.zscaler.com/zia/configuring-sandbox-policy
SSL inspection	Unlimited capacity
File retention	Zscaler Cloud Sandbox operates solely in memory. Files are stripped of identifiable information during analysis. After analysis is completed, benign files are purged from memory while malicious files are encrypted and stored indefinitely, sharing insights across all Zscaler's users for continuous protection (Cloud Effect).
Protocol support	HTTP, HTTPS, FTP, FTP over HTTP
Files per day	Unlimited capacity (inline)
Deployment method	Cloud native, no on-premises hardware or NGFW hooks needed
Threat intel integration	40+ security partner threat intel feeds
Management and reporting	Full reporting including malware behavior and intent, indicators of compromise (IOCs), dropped files, PCAPs
Forensics	Initial sample, secondary payloads, PCAPs
API support	Robust API support, two sample submission APIs and report retrieval via API in JSON format in summary or detailed formats
Granular policies	Easy to use and configure policies for users, location, location groups, file types, user groups, departments, URL categories, and protocols
3rd Party Independent Validation	AAA rating by CyberRatings

2.4.3. Zscaler Sandboxing Licensing

Zscaler is proposing the advanced sandboxing license for Higher Colleges of Technology (HCT)

Feature	Value	Standard Cloud Sandbox	Advanced Cloud Sandbox
File type support	Broad file-type coverage. Expand protection against evolving threats	 - Windows Executables - Windows Libraries - Zip archives	 Check the following documentation for the list of supported file types: https://help.zscaler.com/zia/configuring-sandbox-policy
AI Instant Verdict (AI/ML Quarantine and Blocking)	Allows benign or blocks malicious files in seconds with high confidence		 Available for select file types
Granular policies	Tailor security policies using multiple criteria--users, locations, URL categories, and more	–	 Granular policies fitting many use cases can be created with ease
Reporting	Detailed reports with MITRE ATT&CK matrix mappings, TTPs, malware behavior and intent, IOCs, PCAPs, and more for actionable insights	–	 Multiple reports and dashboards are available, provides rich insights
API	Out-of-band API for on demand Sandbox file analysis	–	 Advanced Sandbox Submission API* (includes dynamic analysis) includes 3000 files per month by default Deep Inspection Scan API* (not including dynamic analysis) includes 5 files per day *Additional licensing may be required based on API usage requirements
Zero Trust Browser Integration	Automatically quarantine risky files allowing users to safely access the file during sandbox analysis	–	 Available for PDF and Microsoft Office files

2.5. Zscaler Zero Trust Browser (ZTBrowser)

Zero Trust Browser provides an organization the capability to isolate users from potentially harmful content on the internet. This is done by loading the accessed web page on a remote browser in any one of the many Zscaler data centers across the globe and streaming the rendered content as a stream of pixels to the user's native browser.

The overall solution of Zero Trust Browser consists of three primary components:

- A browser extension that handles web-based security and access
- A lightweight agent that enforces device posture controls and advanced data protection during the session
- And a cloud browser for accessing private applications in an isolated session.

Isolating web pages on an ephemeral, remote browser ensures that the HTML files, CSS files, JavaScript files, and any other active content served by the accessed web page never reach the user's machine or the corporate network, thus ensuring an air gap between the user and the web page they access.

Zero Trust Browser not only provides the capability to isolate web pages but also allows the user to view file types in isolation without requiring a download of the files to their local machine.

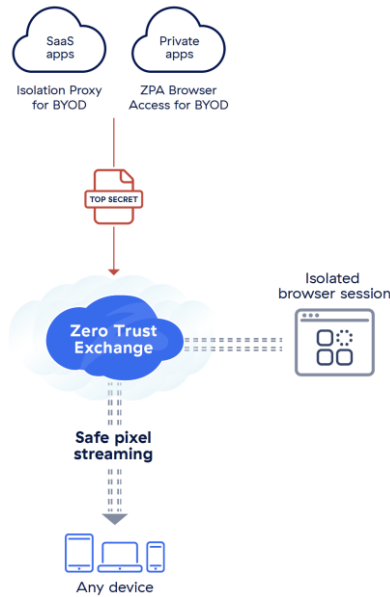
2.5.1. Zscaler Cloud Browser Key Capabilities

- **Secure Browsing:** Air-gapped browsing stops web threats to complement URL filtering, including AI-driven threat isolation stopping threats in any site that presents risk.
- File isolation balances security and productivity, allowing users to interact with files in isolated sessions while preventing file-based threats from taking hold.
- **Secure Agentless Application Access:** Allow employees, contractors and third parties to safely access and use applications from any unmanaged device without the need for an agent. Allow employees, contractors and third parties to safely access and use applications from any unmanaged device without the need for an agent or VDI.
- Fully isolate applications from users to eliminate the risk of vulnerable clients and malware-infected endpoints being leveraged by attackers to compromise applications.
- **Stops Data Leakage:** Allow secure access to web-based SaaS and private applications while restricting copy, paste, and print to prevent data leakage and theft. Get granular control of upload and download activity across SaaS and private applications to protect confidential business data from ending up on unmanaged endpoints.
- In addition to browser-based control, Zero Trust Browser enjoys full integration with Zscaler Data Protection portfolio to protect against sensitive data leakage.

2.5.2. How Zscaler Cloud Browser Works

For managed devices with Zscaler Client Connector, triggered policies for any app or web destination automatically isolate the session.

For unmanaged devices without ZCC, agentless access occurs via browser-based access in the Zero Trust Browser for both SaaS and web apps.



2.5.3. Zscaler Zero Trust Enterprise Browser (EB)

Zero Trust Enterprise Browser for Zero Trust Browser is for accessing private applications while in an isolated session. Traditional security technologies are not designed to focus on securing daily browser execution, and they often lack the needed layers of protection for all the complex operations that browsers perform. This feature enables organizations to detect, investigate, and control browser-native threats and sensitive data interactions in real time.

Enterprise Browser protects organizations without relying solely on traditional endpoint or network-centric controls. Instead, it combines several proactive methods: threat prevention, data encryption, and policy enforcement. It also utilizes tamper-resistant runtime on the browser for the entirety of a user's isolated session. Additionally, it continuously monitors malicious actions occurring inside browser sessions and fully analyzes user behavior as soon as the user starts interacting with web content.

The Enterprise Browser component prevents common attacks on users that occur in daily operations within browsers, such as:

- Accessing SaaS applications
- Uploading or downloading files
- Using browser extensions
- Interacting with generative AI (Gen AI) platforms
- Authenticating to cloud services
- Browsing internal and external websites
- Sharing sensitive information through collaboration tools

This feature is automatically a part of the Zscaler offerings for Zero Trust Browser, and admins do not have to manually enable it for users. When a user enters isolation, the isolated browser utilizes the Enterprise Browser capabilities and maintains the native look and feel of a regular browser.

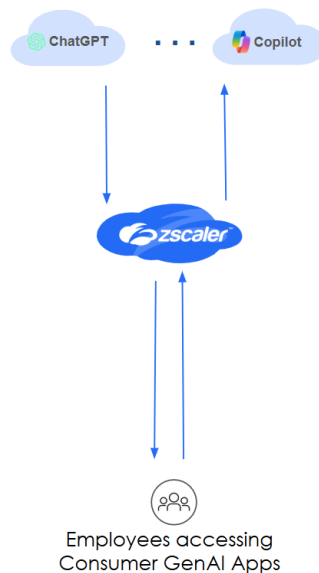
2.6. Zscaler AI Guard for Users

The rise of generative large language models (LLMs) has accelerated the pace of application innovation and increased productivity, with organizations customizing LLMs to their specific tasks. However, the proliferation of LLMs in apps and agents is creating new security challenges, including prompt injection, toxicity, and code leakage. This increasing adoption of LLMs necessitates the implementation of AI security in applications, in addition to cybersecurity and data security.

AI Guard is a service that provides run-time protection for your AI applications by enforcing enterprise policies that prevent prompt injections, block jailbreak attempts, and stop personal information leakage with guardrails around LLM interactions.

It secures these interactions by filtering harmful or inappropriate content from the prompts that users enter and the responses they receive in the AI application. AI Guard enforces intent-based detectors on both prompts and responses.

Secure Access to AI



2.6.1. Key Capabilities of AI Guard

The following functionality and guardrails are some of the options available to you with AI Guard:

- **Visibility & Access Control:** You can monitor which applications are accessing private AI systems, with granular controls over access permissions. Visibility tools ensure complete oversight of private AI interactions across the enterprise.
- **Prompt Injection & Jailbreak Protection:** AI models can be manipulated with adversarial prompts to bypass restrictions or produce unintended outputs. The AI Guard guardrails detect and neutralize such attacks, ensuring that AI responses stay within defined safety parameters.
- **Toxicity:** AI-generated content can potentially be emotionally harmful for workers. AI Guard's guardrails apply real-time monitoring and filtering to prevent toxic language.
- **Sensitive Data Protection:** AI models can inadvertently expose or process confidential, regulated, or proprietary information. AI Guard enables inline inspection and classification to detect and block sensitive data before it is sent to AI APIs.
- **Off-Topic Response Detection:** AI Guard detects and prevents off-topic responses, ensuring that AI stays focused on its intended use cases.
- **Malicious URL Detection & Blocking:** AI-generated responses can sometimes include harmful or phishing links. AI Guard's solution actively scans links and blocks malicious URLs to prevent security threats and safeguard users from cyber risks.
- **Language Detection & Enforcement:** Organizations can enforce AI interactions in specific approved languages, preventing unauthorized language use that can lead to miscommunication or security vulnerabilities.
- **Code Injection & Execution Detection:** AI Guard detects and prevents unauthorized code snippets embedded in AI interactions, protecting against potential security risks from malicious scripts or unauthorized automation attempts.

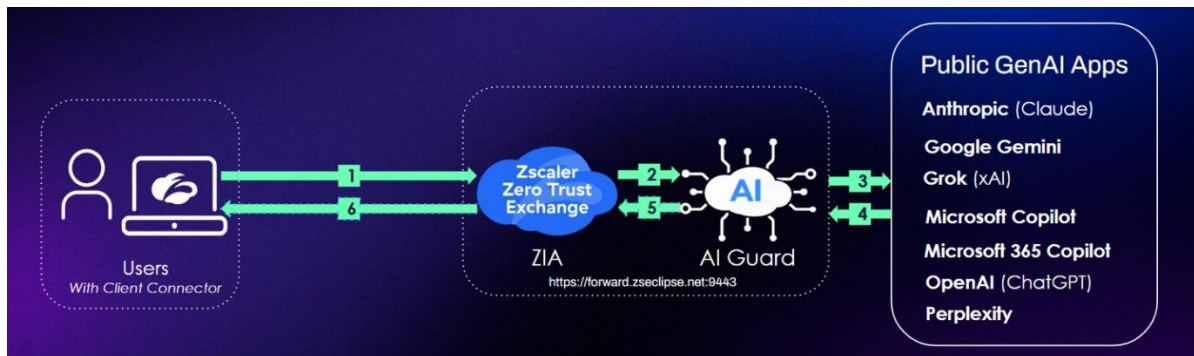
- **Gibberish & Low-Quality Prompt/Response Filtering:** AI models can sometimes produce nonsensical or low-value responses. Our system detects gibberish, meaningless text, and irrelevant output, ensuring that AI responses remain coherent and valuable.
- **Refusal Detection & Intervention:** When AI refuses to respond to valid queries due to overly restrictive filters or misunderstanding, our system flags and intervenes to alert the security team because it might be a denial-of-service attack.
- **Finance Advice:** AI Guard can block actionable financial guidance (investing, trading, tax, product choices); allows neutral finance facts, history, and definitions.
- **Prompt Tagging-based Access Control:** AI Guard has prompt tagging model that automatically classifies prompts into different categories, enabling organizations to enforce AI governance, security, and compliance while optimizing AI interactions.
- **Competitor Discussion Detection:** You can identify and block prompts that reference competitors, rival products, pricing comparisons, or market strategies.
- **URL Reachability Detection:** AI-provided links can sometimes be outdated or unreachable. This guardrail verifies whether URLs in prompts are accessible, safe, or broken to prevent hallucinations.
- **Legal Advice:** Block prompts seeking legal advice, interpretation, or compliance guidance. This allows for neutral legal facts, definitions, and non-legal work inquiries.

2.6.2. AI Guard for Users Architecture

AI Guard for Users is a service that provides run-time protection for your generative AI (GenAI) applications on a per-user or user group basis by enforcing enterprise policies that prevent prompt injections, block jailbreak attempts, and stop personal information leakage with guardrails around LLM interactions.

It secures these interactions by filtering harmful or inappropriate content from the prompts that users enter and the responses they receive in the AI application. AI Guard enforces intent-based detectors on both prompts and responses.

The AI Guard for Users architecture includes the following key components:



1. **Signing in to ZIA:** Zscaler Client Connector authenticated users connect to ZIA in order to send traffic to it.
2. **User prompt is analyzed by AI Guard:** ZIA forwards traffic to the public GenAI apps using proxy chaining to the AI Guard infrastructure within Zscaler cloud (traditional Cloud App Controls, Web DLP, Threat Protection still occur here). If an AI Guard policy blocks the prompt, the traffic will not go to the GenAI apps.
3. **Prompts that aren't blocked are sent to GenAI apps:** If the prompt isn't blocked, AI Guard initiates (proxies) the traffic to the Gen AI apps.
4. **GenAI apps response sent to AI Guard:** The public GenAI apps return their response to AI Guard, where it is analyzed with policy response detectors if configured. If blocked, the user will instead receive a blocked message from AI Guard.
5. **Response sent through ZIA:** The response is sent back to ZIA.
6. **User receives response:** ZIA returns the response to the user.

3. Proposed Solution: Zscaler Private Access (ZPA)

The Zscaler Private Access (ZPA) service enables organizations to provide access to internal applications and services while ensuring the security of their networks. ZPA is an easier to deploy, more cost-effective, and more secure alternative to VPNs. Unlike VPNs, which require users to connect to your network to access your enterprise applications, ZPA allows you to give users policy-based secure access only to the internal apps they need to get their work done. With ZPA, application access does not require network access.

While ZPA is for connecting users to an enterprise's internal applications, Zscaler Internet Access (ZIA) is for connecting users to public applications on the internet.

Additionally, ZPA decouples applications from the physical network so you can provide seamless connectivity to private internal applications and assets whether they are in the cloud, the data center, or both. It also adjusts dynamically to network changes, so you can move your resources without impacting user access.

You can configure settings and policies on a central ZPA Admin Portal, which also feature dashboards where you can see your users and the apps they access, and monitor the health of your servers and resources. You can configure ZPA to automatically discover servers and applications when users request them, or you can configure them manually as well. You then define policies that specify which apps users or groups can use, and ZPA allows them to connect to those apps only. ZPA renders your applications invisible to all but the authorized users and unrouteable to anyone.

3.1. Key Capabilities

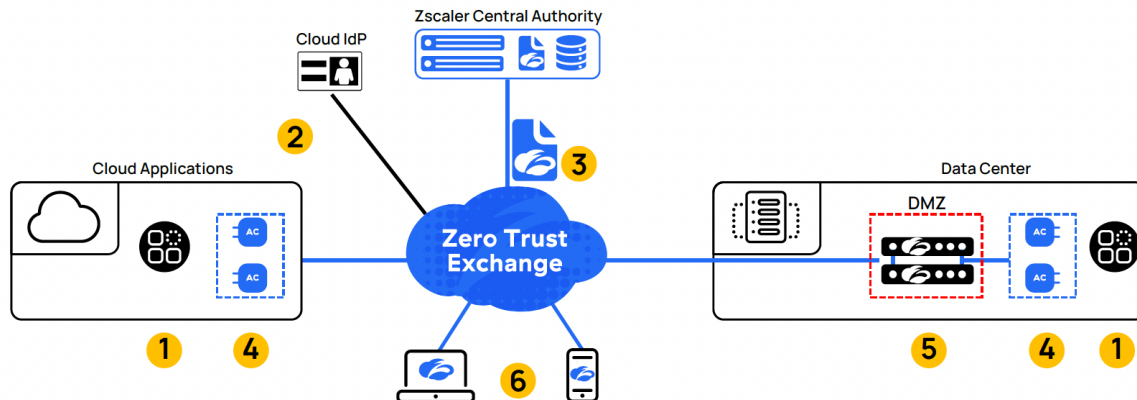
Below are ZPA key features and benefits:

- **Seamless User Experience:** Policy-driven connectivity that dynamically adjusts to network changes.
- **Enhanced Security:** Application-specific connectivity without ever bringing users on-net.
- **Ease of Deployment:** Does not require hardware or hardware upgrades.
- **Instant Deployment and Discovery:** Can automatically discover applications so you can easily build policies around them.
- **Single Sign-On (SSO):** ZPA is tied directly to your existing authentication infrastructure, leveraging SSO to further reduce complexity.
- **Real-Time Visibility:** Dashboards provide unparalleled visibility into your users and applications, and the health of your organization's applications and servers

3.2. Getting Started with ZPA

ZPA allows you to define a set of policies, authentication, and application definitions that combine to provide user access to applications. The ZPA service ensures that applications are available and that users have the correct policy applied.

The following ZPA components and configuration elements compose the foundation for implementing zero trust principles via user segmentation.



1. **Applications, Application Segments, and Segment Groups** – An application is a Fully Qualified Domain Name (FQDN), local domain name, or IP address that is defined on a standard set of ports. Applications must be defined within an application segment. Zscaler recommends using FQDN whenever possible. An application segment is a set of defined applications on shared ports across one or more back-end servers. A segment group is a set of application segments combined for policy purposes. Applications can be grouped into application segments and segment groups based on access type, authorized users, etc.
2. **SAML and SCIM Attributes** – SAML and SCIM attributes such as group membership, role, etc. are used in access policy rules to provide least-privilege access to applications. These attributes may originate in existing authentication/ authorization repositories, such as Active Directory, binding users to relevant groups that reflect onboarding, movement to different departments, changes such as termination, etc.
3. **Access Policy** – Access policy rules enable context-based access control. To configure an access policy rule, you must first define which applications or segment groups the rule controls, and then define the context required for access. Additional context for access policies may include device posture, access type, network location, and other context provided by the SAML Identity Provider (IdP).
4. **App Connectors** – App Connectors provide a secure, encrypted, authenticated interface between a customer's servers and applications and the ZPA cloud for delivering user traffic to back-end applications.
5. **Private Service Edge** – A ZPA Private Service Edge is a single-tenant instance that provides complete broker functionality of a ZPA Public Service Edge in an organization's environment. For an on-premises user connecting via a local Private Service Edge and local App Connector, all control-plane and data-plane traffic stays within the network; the Private Service Edge communicates to the Zscaler cloud for management plane (configuration, logging, etc.) and delivery of user traffic to remote resources.
6. **Zscaler Client Connector** – Zscaler Client Connector is an agent that resides on your mobile or desktop devices. Supported on popular operating systems such as Windows, macOS, Android, iOS, and CentOS, this agent connects your devices to Zscaler's Zero Trust Exchange. At the ZPA Service Edge, requests are evaluated and approved users are connected to appropriate applications.

3.3. ZPA Cloud Architecture

ZPA was built from the ground up to provide secure remote access.

ZPA runs on a unique multi-tenant infrastructure, separate from that of Zscaler Internet Access (ZIA) solution, in order to make the service highly available. This separation ensures that when the same organization is using both services, public and internal traffic never mix on the same device. Additional measures are used to ensure security and privacy between organizations using the ZPA service, and in many cases, traversing the same equipment. Finally, all communication is encrypted end-to-end.

The ZPA service includes the following key components and subsystems:

3.3.1. ZPA Central Authority

The ZPA Central Authority (CA) is the "brain and nervous system" of the ZPA cloud. It is a geographically distributed system of in-sync nodes that are all actively servicing requests. To learn more, see ZPA Public Service Edges and the Central Authority.

The ZPA CA monitors the cloud and provides a central location for software and database updates, as well as policy and configuration settings. Each component of the system is aware of the health of the entire cloud, and policy is distributed across all components of the CA. It includes user interfaces connected via APIs to configuration components, as well as distributed dispatchers that track real-time application states.

3.3.2. ZPA Public Service Edges and ZPA Private Service Edges

ZPA Public Service Edges and ZPA Private Service Edges enforce user policies and provide secure transport to App Connectors. Maintained by Zscaler and deployed globally, ZPA Public Service Edges are the cloud-based portion of the ZPA data forwarding path, responsible for giving Zscaler Client Connector and App Connectors the ability to provide secure, authenticated, policy-based access to internal applications. ZPA Private Service Edges act in the same capacity, but they are managed by an organization. The ZPA Public Service Edges and Private Service Edges are responsible for a variety of functions, including:

- Zscaler Client Connector and App Connector configuration and authentication, using public key cryptography.
- Creating and managing M-Tunnels, including authentication, policy, setup, teardown, and data forwarding. M-Tunnels are end-to-end encrypted connections designed to ensure that all traffic stays separate.

3.3.3. App Connectors

App Connectors provide the authenticated secure interface between an organization's application servers and the ZPA cloud. App Connectors can be distributed using lightweight virtual machines (VM) to run in legacy enterprise data centers and public cloud environments or as a package that can be installed under supported operating systems.

App Connectors are deployed in locations where internal applications reside, or they can be deployed in any location that has connectivity to those applications. ZPA selects the closest App Connector given the location of the user and the App Connector-to-application latency. It is best to install them in redundant (N+1) configurations, where (N) is the number of App Connectors required to support the applications, because they are always active. App Connectors never communicate with each other and do not accept inbound connections. They only require an outbound connection to the internet to reach ZPA Public Service Edges or Private Service Edges, so they can provide users access to internal applications in your organization.

Using a key pair (i.e., a provisioning key and a corresponding TLS client certificate), the App Connector and the ZPA cloud verify each other as part of enrolment. After an App Connector is enrolled, the TLS client certificate allows it to maintain its authentication with the nearest ZPA Public Service Edge or Private Service Edge, which provides the App Connector with its configuration and allows it to begin its operational tasks. These tasks include:

- Find applications as requested by users
- Act on behalf of the application to send traffic back to users
- Complete DNS requests on demand, based on established policies
- Monitor applications for their reachability

3.3.4. Logging Streaming Service

The Log Streaming Service (LSS) provides real-time and historical visibility into the operation of ZPA by analyzing events reported, primarily by the ZPA Public Service Edges or Private Service Edges. In addition to real time analytics and status, the LSS provides short and long-term storage.

Information sent to the LSS includes user activity and status, connector status, and http information related to Browser Access. The default fields in log templates created by ZPA do include personally identifiable information (PII). However, your organization can create a custom log template that only captures specific fields.

The LSS has two components: a log receiver and an App Connector. It can stream logs from your ZPA tenant to a SIEM or log receiver you manage. You must specify the logs required for streaming in the ZPA Admin Portal.

Because ZPA was purpose-built to ensure secure, private access to internal applications, the service does not store any information on any type of persistent media (i.e., disk) unless it is configured to do so by your organization. The information can also be obfuscated based on your organization's preferences.

3.4. Traffic Forwarding

There are two ways to manage traffic forwarding:

3.4.1. Zscaler Client Connector

Zscaler Client Connector is a client or app that is installed on a user's device (or endpoint), which authenticates the endpoint with the ZPA cloud and enables access to internal applications on behalf of that endpoint. Zscaler Client Connector verifies the integrity of both the ZPA infrastructure and the peers it is communicating with using public key cryptography.

Beyond the full authentication, as handled as part of the standard SAML 2.0 workflow, Zscaler Client Connector creates a unique fingerprint of the user's device. This means if someone copies Zscaler Client Connector from one device to another, the fingerprint generated by Zscaler Client Connector will prevent successful enrollment from the new device.

Zscaler Client Connector recognizes the internal applications that are available via ZPA. Access to these applications is defined in ZPA based on policies. Using information received from the ZPA Public Service Edge or ZPA Private Service Edge, it intercepts the user requests for those applications and then forwards those requests to the ZPA cloud. No network information is required to access available applications, and an organization can specify if resources must go through ZPA at all times, when on an untrusted network only, or never.

To facilitate secure private connections that are abstracted from the physical network, Zscaler Client Connector associates permitted internal applications with a set of synthetic IP addresses. So, when a client application (i.e., a web browser or SSH client) sends out a DNS request, Zscaler Client Connector is able to recognize the domain as an internal application being protected by ZPA. Zscaler Client Connector then intercepts the DNS request and delivers a DNS response to the client application that uses the synthetic IP address associated with the internal application.

Organizations can also use Zscaler Client Connector to direct user traffic to ZIA. Zscaler Client Connector identifies internal applications to forward traffic to ZPA, and other traffic goes to ZIA. If an organization is using ZIA and ZPA, Zscaler Client Connector will inherit some functionality from ZIA, including enrolment, user profiles, traffic management, etc. However, organizations do not need access to ZIA in order to use ZPA.

3.4.2. Browser Access

Browser Access allows you to leverage a web browser for user authentication and application access over ZPA. This allows users who cannot install the Zscaler Client Connector on their devices, but who still need access to applications, to do it over ZPA.

The application must be available via HTTP/HTTPS in a web browser that supports TLS 1.2. When a user tries to access an internal web application, they can open any web browser on their device and enter the application's URL. ZPA directs this traffic to the organization's identity provider (IdP), where the IdP authenticates the user and grants a SAML certificate before forwarding the traffic to a ZPA Public Service Edge or Private Service Edge. Browser Access is session based, so the user must authenticate via the IdP every time they access try to access the web application.

The ZPA Public Service Edge or Private Service Edge accepts the SAML certificate and applies all policies, making only permitted internal web applications accessible to the user. If the policy allows the user access, ZPA identifies the best-path App Connector for the internal web application and connects them to it. When the ZPA Public Service Edge or Private Service Edge sends out a request to all App Connector sitting at the application level, the App Connector sitting closest to the internal web application responds by connecting to the ZPA Public Service Edge or Private Service Edge.

The connections from user to ZPA Public Service Edge (or Private Service Edge) and App Connector to ZPA Public Service Edge (or Private Service Edge) are then stitched together forming a TLS-encrypted tunnel, creating a secure channel between the authorized user and the internal web application without granting network access.

3.5. Understanding Communication Between Components and Subsystems

All communication between ZPA components travel within a mutually pinned, client and server certificate-verified, TLS connection. Within this TLS-encrypted Zscaler Tunnel, a microtunneling protocol exists. Select components of ZPA run through this encrypted Microtunnel end-to-end.

Because client and server use pinned certificates, it is cryptographically impossible for ZPA to experience a Man-in-the-Middle (MITM) attack. The client certificates are verified against an organization's Certificate Authority (CA), and the server certificates are verified against Zscaler's CA, which cannot be spoofed by any third-party compromised CA.

All Zscaler certificates are signed through an offline-only, air gapped, signing ceremony. ZPA only accepts connections from the Zscaler Client Connector and the App Connector instances that present a client certificate signed by a CA associated with each tenant. Zscaler Client Connector and App Connector will only connect to ZPA service components that present a certificate signed by the ZPA infrastructure PKI.

All certificates used within the ZPA infrastructure, including all organization-operated systems, have private keys that never leave the physical device in which they were generated. The private keys for each organization's Central Authority (CA) are stored in encrypted format (i.e., AES256-GCM). Proper best practices for public key lifecycles are always maintained.

3.5.1. Zscaler Tunnel

A Zscaler Tunnel (Z-Tunnel) is a TLS-encrypted, mutually authenticated point-to-point connection between Zscaler Client Connector and a ZPA Public Service Edge managed by Zscaler, or it's between an App Connector and a ZPA Private Service Edge managed by an organization. A Z-Tunnel does not contain any direct IP data. An additional Z-Tunnel is created between Zscaler Client Connector and a ZPA Public Service Edge, or between an App Connector and a ZPA Private Service Edge, to support multi-tenancy in Zscaler Client Connector. This allows Zscaler Client Connector to establish a Z-Tunnel with the tenant and establish a separate Z-Tunnel with the partner tenant. The second Z-Tunnel works without any additional setup or configuration. Also, the Z-Tunnel can carry multiple communication channels called Microtunnels.

Z-Tunnels are authenticated using public key cryptography. Zscaler Client Connector and App Connectors are authenticated with the organization's public key infrastructure (PKI). ZPA Public Service Edges and Private Service Edges are authenticated with Zscaler's PKI. In addition to the certificate-based authentication already applied, the Z-Tunnel is further authenticated using SAML 2.0.

3.5.2. Microtunnel

A Microtunnel (M-Tunnel) is an end-to-end communication channel created between Zscaler Client Connector and an internal application via a ZPA Public Service Edge or Private Service Edge and an App Connector upon demand.

An M-Tunnel originates at Zscaler Client Connector when traffic from a client application (i.e., a web browser or SSH client) attempts to connect to an application server, which is associated with a synthetic IP address. The M-Tunnel extends to the ZPA Public Service Edge or ZPA Private Service Edge using a tag created by Zscaler Client

Connector. The M-Tunnel then extends to the appropriate App Connector using a different tag chosen by that connector. Finally, the App Connector connects the M-Tunnel to the actual application server providing the service originally requested by the client application.

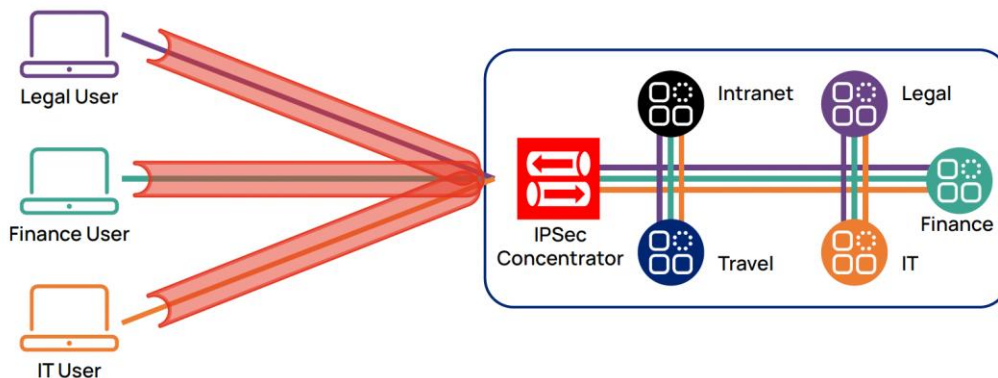
At no point in the communication is actual IP identification data passed. Instead, the combination of tags created by Zscaler Client Connector and the App Connector function similarly to the label stack in a Multiprotocol Label Switching (MPLS) network. Just as in an MPLS network, where the label-switched path (LSP) uniquely identifies a connection, Zscaler Client Connector and App Connector tags identify a unique connection for each application request.

3.6. Transitioning to a Zero Trust ZPA Deployment

One common driver for ZPA deployment is to replace a remote access IPSec VPN. In the VPN model, endpoints are connected to a network, and access security must be handled by network segmentation or application controls. In your initial deployment of ZPA, you can do something similar with wildcard discovery. You can deploy ZPA in a policy configuration that mirrors the open-access VPN model with wildcard app segments and broad access policies. This allows you to make one conceptual change at a time: first change the transport to the applications, then work on granularity of control. This also reduces risk by ensuring that no user applications break due to policy restrictions.

At a high level, the starting point will look something like the following illustration, where users have access to various applications.

IPSec Clients



When transitioning to ZPA for access, it should be as transparent as possible to the end user. By setting up ZPA to use wildcard discovery, you are replicating the user's existing controls. After migration to Zscaler, initial setup will look something like the following illustration, mirroring your IPSec VPN access policy so users have a similar experience and access to the applications.



After your initial group is migrated to ZPA, you need to begin building more granular policy. The { the app segment definitions, user context, and access policies based on what is learned in discovery, and to

provide access to required applications only to users and user groups who should have access to these resources. This can be achieved in

multiple ways depending on the requirements. The same process of discovery and refinement allows you to extend your ZPA protection beyond the remote-access use case and also encompass on-premises user access.



How your organization progresses toward a zero-trust transformation will be defined by the kinds of access you want to grant and the relationship of the user, device, or location to your organization. Some use cases require using fully granular zero trust policies. Examples include third-party access to internal tools, or during an acquisition where the user

community is easily identified and the target resources are clearly scoped.

For other use cases, such as employee roles where the traditional approach has been full network access, a different approach may be required. A targeted access policy can allow only those users to get to a broad range of resources (e.g., *.internal.safemarch.com), enabling initial connectivity as well as dynamic application discovery. The resulting user activity logs offer visibility to inform development of more granular policies, so you can evolve towards more granular zero trust controls for those users as context becomes available.

It's important to remember that "success" at zero trust does not necessarily mean eliminating all wildcard access. There may be use cases in your organization where it's appropriate to provide a certain user community access to a full range of

applications within a certain domain or subdomain. Granularity then might mean ensuring that only authorized users on appropriate devices, possibly with additional context controls, are permitted access via that wildcard policy.

3.7. AI-Powered Segmentation

Zscaler's AI-powered Intelligent Application Segmentation is an advanced capability that streamlines the Zero Trust journey. It delivers precise user-to-Application Segmentation and tailored Access Policy recommendations, ensuring secure and efficient connectivity for users across the organization.

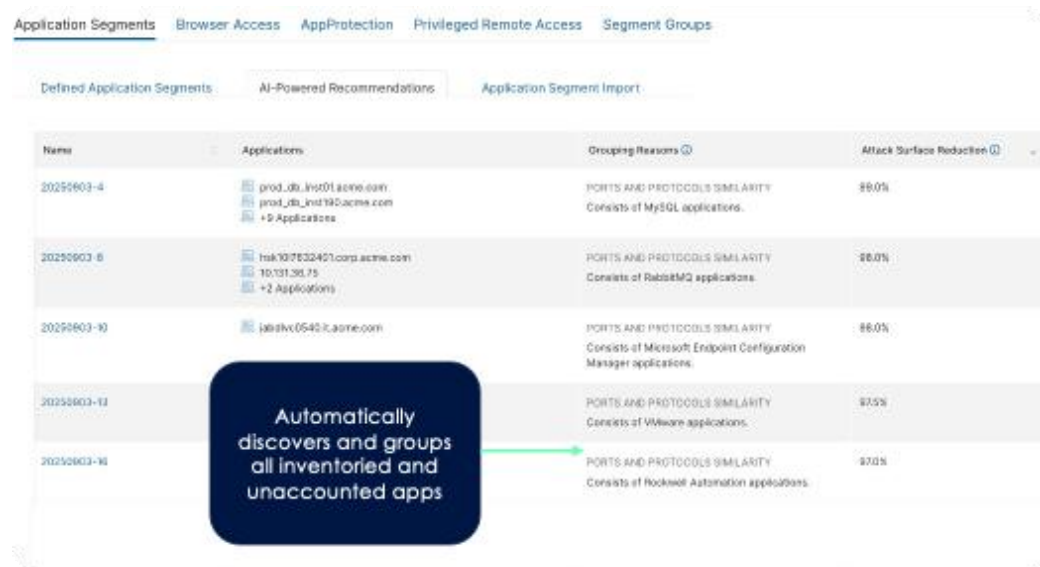
The AI/ML-powered Intelligent Application Segmentation is a ZPA feature built to assist the ZPA administrator with their journey towards Application Segmentation, providing Zero Trust access for private applications. It performs the following:

- Digests application access logs for the last 30 days
- Using the access patterns of the discovered applications, AI/ML provides recommendations for the following areas:
 - Identifies Application Segments required to serve the observed application requests, which would eventually help with removal of "open" (wildcard) Application Segments
 - Users and/or probable SCIM or SAML Departments/Groups that require the access which can be used in Access Policy definition

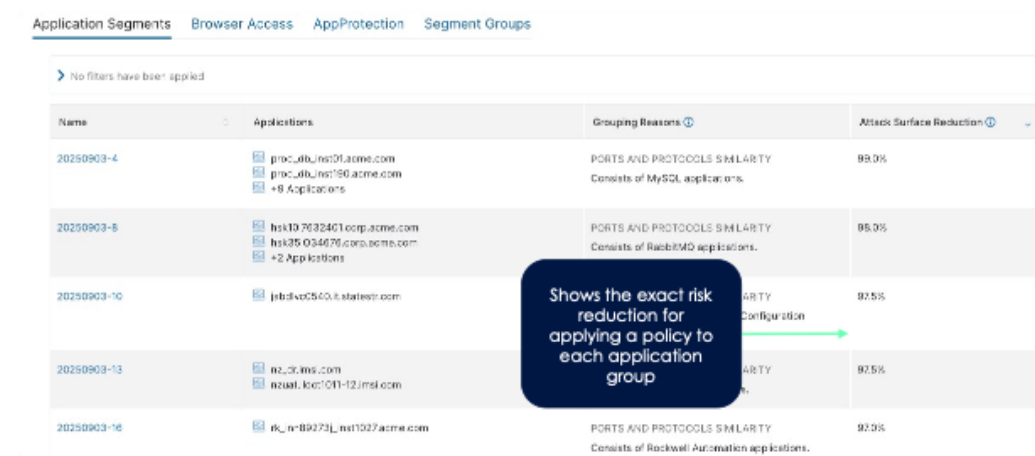
The outcome helps in reducing the attack surface (and henceforth limit lateral movement) by enabling the following:

- Application Segments defined with right set of FQDN/IP(s), port(s) and protocol(s)
- Least privilege access, so only the correct set of users have access to the correct set of applications via Access Policies

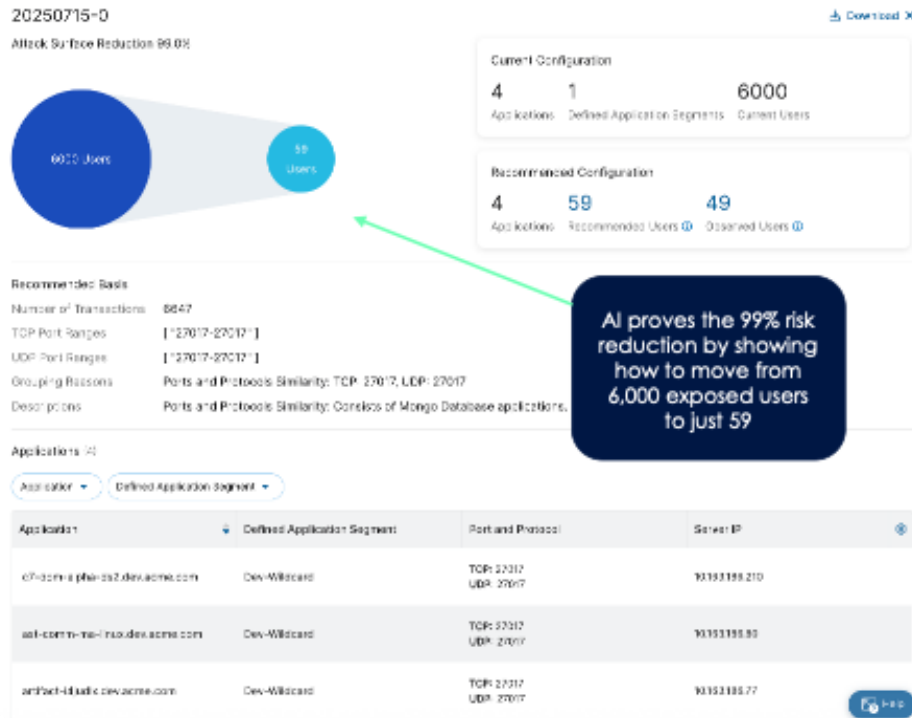
The AI-Powered Segmentation Automatically Discover and Group Unaccounted Apps



The AI-Powered Segmentation shows the exact risk reduction to each application group



The AI-Powered Segmentation details the relevant users for each application and with one-click convert any AI recommendation into a secure application segment



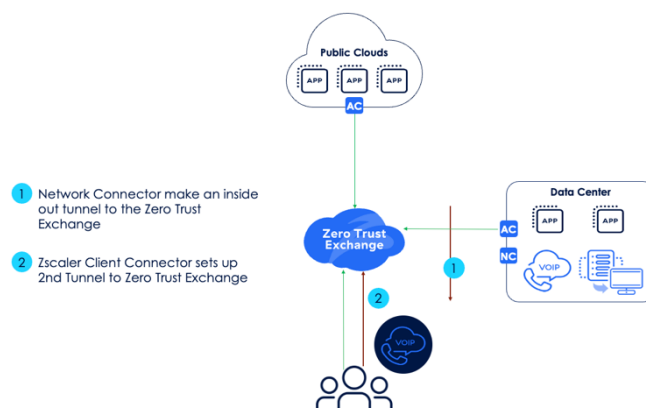
3.8. Zscaler VPN (For Legacy Apps)

The VPN service (for legacy apps) designed to manage exception traffic, specifically for VoIP and S2C applications, while facilitating the transition to a zero-trust network architecture.

This solution ensures continued access to applications that are not immediately adaptable to zero trust access, thereby minimizing disruption during the migration process to critical applications.

3.8.1. ZPA Architecture for VoIP and Server-to-Client Apps: Network Connectors, Secondary Tunnel, and Zero Trust Exchange

High-level solution design:



- Network Connectors (NC) can be deployed in datacenter/campus locations hosting VoIP servers and other server-to-client (S2C) systems. These connectors initiate an outbound connection to the ZTE, establishing a secure tunnel between the connector and the ZTE.
- Zscaler Client Connector (ZCC) builds a secondary tunnel specifically for users authorized for VPN services, also connecting to the ZTE. The ZCC is configured to forward traffic through this alternative path based on the configuration parameters set by the administrator.
- ZTE integrates the connection between the ZCC and the NC, creating an end-to-end encrypted pathway that facilitates secure communication for these applications.

4. Proposed Solution: Zscaler Digital Experience Monitoring (ZDX)

As your applications and work locations continue to evolve, so must your ability to monitor and diagnose application access and performance. Traditional monitoring tools for application performance and user experience relied on the ability to control and instrument the network and applications.

The rapid adoption of cloud and mobility initiatives within organizations and a shift to work-from- anywhere have introduced new monitoring challenges for IT teams. Applications are moving out of the data center and into the cloud.

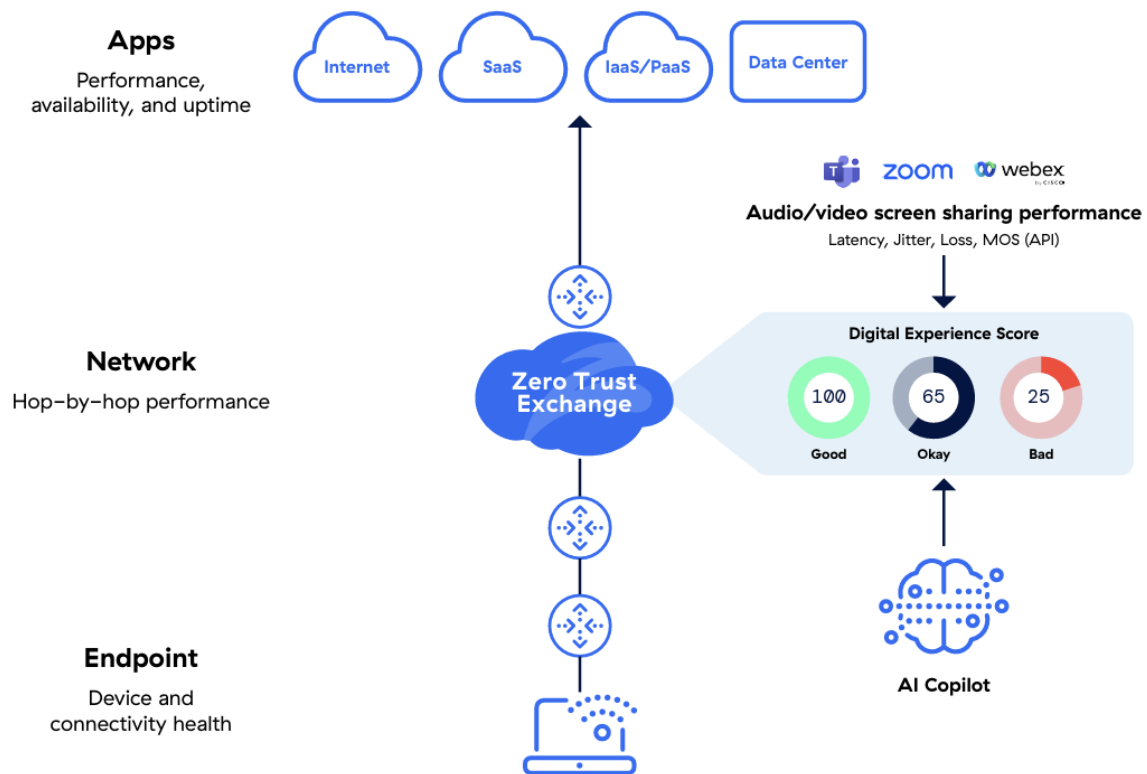
They are being accessed by a hybrid remote workforce, meaning IT teams no longer control the underlying infrastructure and technology stack, and lose end-to-end visibility into the user experience. End user performance issues arising from SaaS or cloud application availability, home Wi-Fi issues, network path outages, or network congestion are not easily isolated and diagnosed.

Digital experience monitoring for a hybrid workforce requires a modern and dynamic approach. IT teams need to continuously monitor and measure the digital experience for each user from the user perspective, regardless of their location. Traditional monitoring tools take a data center-centric approach to monitoring and collecting metrics from fixed sites rather than directly from the user device. This approach does not provide a unified view of performance based on a user device, network path, or application.

Zscaler Digital Experience (ZDX) is an AI-powered digital experience monitoring solution delivered as a service from the Zscaler cloud. ZDX provides end-to-end visibility and troubleshooting of end or customer performance issues for any user or application, regardless of location.

In addition, it enables continuous monitoring for network, security, and service desk teams with insight into the end user device, network, and application performance issues.

ZDX leverages Zscaler Client Connector and the Zscaler Zero Trust Exchange to actively monitor applications from an end user or customer perspective. It continuously collects and analyzes various performance metrics, including application availability, response times, network hop-by-hop performance metrics, and device health metrics such as device configuration, CPU, memory usage, process information, and device events. As a result, network operations and service desk teams get uninterrupted visibility and save time with proactive identification and resolution of end user experience and application or services issues.



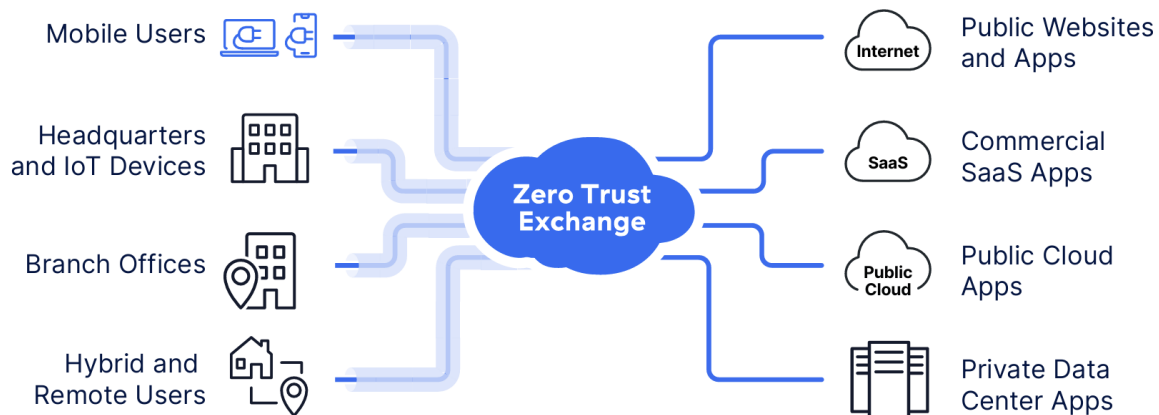
4.1. Key Capabilities

ZDX lets you monitor granular details for users, devices, applications, and more in your org. With ZDX, you can:

- Proactively monitor every user device in your organization to detect user experience and productivity issues.
- Grade your digital experience by referencing the ZDX Score over time for your organization and each application, location, department, and user.
- Triage performance issues and pinpoint whether the issue is in the local network, a user's device, the ISP provider, or within the Zscaler cloud.
- Use Zscaler's unified client, Zscaler Client Connector, to implement ZDX on Windows, macOS, Android, Android on ChromeOS, and iOS devices.
- Monitor general data center information and your key SaaS applications (i.e., Outlook Online, Salesforce, Box, etc.) using HTTP, ICMP, or UDP synthetic probes running on user devices in near real time.
- Collect real-time device health information (i.e., CPU usage, memory usage, Network IO, Disk IO, Wi-Fi signal strength, etc.) for your user devices.
- Hop-by-Hop network path visualization from the client device to the destination device.
- Threshold-based alerting for SLA monitoring.
- Remote troubleshooting capabilities to perform detailed analyses of problematic devices (i.e., system process-level monitoring, full network path information including the Autonomous System Number (ASN), ISP information, country hop information, and packet tracing).
- Executive insights reports and digital experience analytics aggregated at organization, location, and application levels.

4.2. Understanding ZDX Probes

For many years, traditional network monitoring tools existed with cloud applications by monitoring connections from fixed sites. Remote work was limited and often entailed a VPN connection to a central site for access to applications. The realities of newly remote and hybrid work have made that monitoring style of limited value.



ZDX instead monitors application availability from every user's device via Zscaler Client Connector. This is the same software that allows users to connect directly to applications using ZPA, or to a ZIA Public Service Edge for internet access. The ZDX feature of Zscaler Client Connector enables probes to be sent to configured web-based applications.

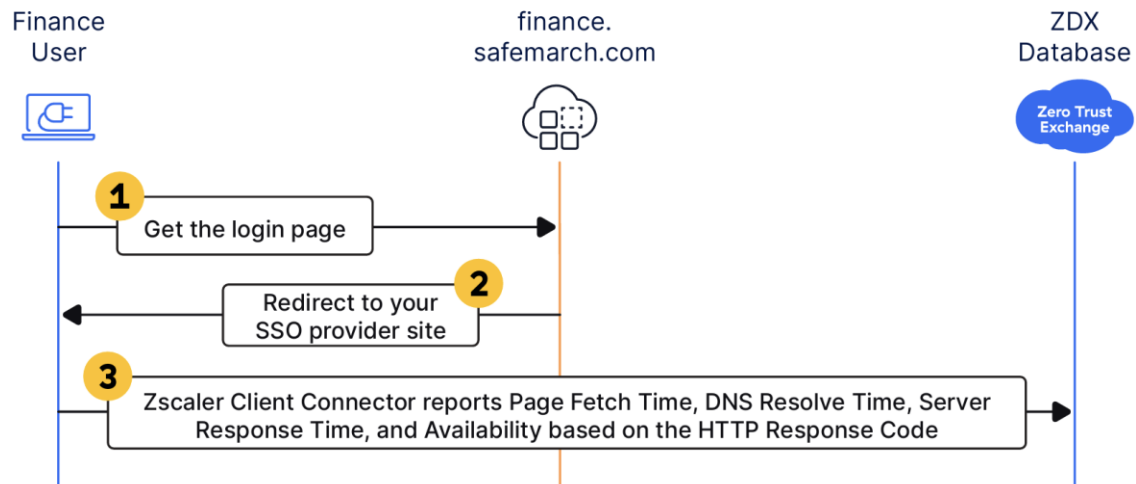
ZDX probes exist as part of an application you select. ZDX provides several preconfigured applications depending on your subscription level, and you can always define custom applications and probes. The probes check that your application is responding appropriately, and record latency and jitter statistics along the network path. These two functions are handled using two probe types:

- Web probes that check for application availability.
- Cloud Path probes that record hop-by-hop network performance information between the user and the application.

The data from these probes is combined with information about the user's device. This includes items such as CPU and memory usage, network usage, and Wi-Fi signal strength. This information is sent to the ZDX cloud. The data is aggregated and presented as a series of dashboards that allow you to see your organization's application performance. ZDX also supports configurable alerts, allowing your operations team to proactively address issues as they begin, not after they are reported.

4.2.1. ZDX Web Probes

The ZDX web probe makes an HTTP or HTTPS GET request to an application to check its service availability and responsiveness. When the application receives the request, it responds with an HTTP Status Code, such as a redirect to a login page. ZDX sends the probe and looks for the correct response code in return based on its configuration to mark the status as available or down.



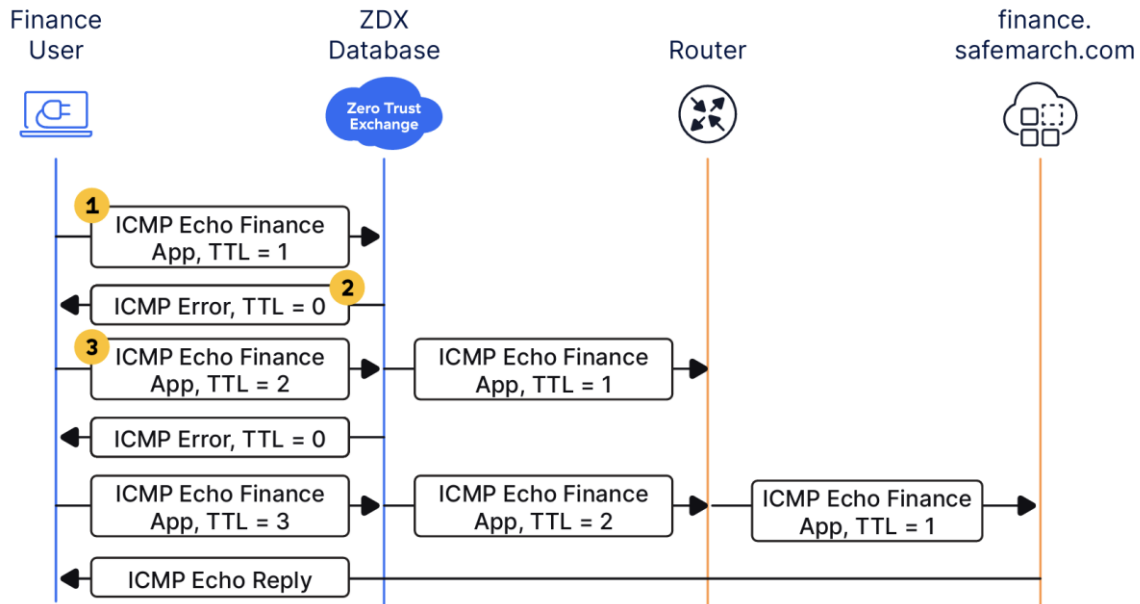
1. A member of the finance team has Zscaler Client Connector installed. Zscaler Client Connector is configured to check that the finance application is running for anyone in the finance group. To do this, Zscaler Client Connector sends an HTTPS probe to the login page for finance.safemarch.com.
2. Finance.safemarch.com responds with a redirect to the organization's single sign-on (SSO) provider to authenticate.
3. The server response and the data around the response is sent to ZDX, which is configured to consider the redirect response as a sign that the service is operational.

The ZDX web probes collect the following information about the application:

- **Page Fetch Time** – This metric collects the network fetch time of the web page from the URL-specified web probe. It requests only the top-level page document and does not request all embedded links within the web page.
- **DNS Resolve Time** – This metric represents the time it took to resolve the DNS name for the hostname specified in the web probe URL.
- **Server Response Time** – Time to First Byte (TTFB).
- **Availability based on the HTTP Response code** – If a success code is returned, the availability is either 1 or 0. If the probe times out, the availability defaults to 0.

4.2.2. ZDX Cloud Path Probes

Cloud Path probes work by sending packets with reduced time-to-live (TTL) values to map the path between the user-installed Zscaler Client Connector and the web application. Cloud Path probes can use ICMP, UDP, or TCP, and can change these segment by segment. The system records the statistics as the probes come back or fail to come back.



1. The same finance user with Zscaler Client Connector installed, the agent begins sending probe packets out with a time to live (TTL) of 1, or 1 network hop away. The probes are sent 5 times with the same TTL.
2. At each hop, the router decrements the TTL by 1, dropping any probes that reach 0. Typically, a router sends a notification back to the sender that the TTL has expired, and that the packet has been dropped. Zscaler Client Connector records this data for each set of probes.
3. After the 5 sets of packets, Zscaler Client Connector increases the Cloud Path probe TTL by 1 and sends another series of 5 probes. This increase continues until the application is reached or the number of hops exceeds the systems limits.

Cloud Path probes are used to collect the following metrics:

- **Hop Count** – The number of hops between each hop point on the path.
- **Packet Loss** – The percentage of packet loss at each hop point on the path.
- **Latency (Average, Minimum, Maximum, and Standard Deviation)** – The roundtrip path time measured in milliseconds.

4.3. Applications Monitoring

Application definitions allow you to build web and Cloud Path probes to the application for monitoring. ZDX comes with a set of predefined applications as well as the ability to build custom applications. When using a predefined application, you enable the application and enter your tenant information for your organization. With a custom application, you need to build the web and Cloud Path probes to support that application.

Configuration

End User

20

Top Private Apps

Search

+ Add

Unified Communication ...

Predefined Apps Collecti...

Custom Apps Collection

Predefined Apps Collection

Enabled

Total: 16 Probes: 8 8

Collection Details

View Details

Status

Search

Add Application

Application Name	Status	No. Probes	Last Updated	Actions
Box	Enabled	1 1	Jun 20, 2025 02:18 PM GMT+4	
Microsoft Login	Enabled	1 1	Jun 20, 2025 01:22 PM GMT+4	
Microsoft Teams Web App	Enabled	1 1	Jun 20, 2025 01:22 PM GMT+4	
Okta	Enabled	1 1	May 21, 2025 12:35 AM GMT+4	
OneDrive for Business	Disabled	1 1	Jun 20, 2025 02:13 PM GMT+4	
Outlook Online	Enabled	1 1	Jun 20, 2025 02:03 PM GMT+4	

4.3.1. Applications Definition

ZDX has predefined support for common large enterprise applications. Because these applications are already understood by ZDX, you only need to do minimal configuration before using them. The exact applications that are available to your organization depend on your ZDX subscription level. The following list includes all supported applications:

- Unified Communication
 - Microsoft Teams Call Quality
 - Webex Call Quality
 - Zoom Call Quality
- Web Applications
 - Box
 - Microsoft Login
 - Microsoft Teams Web App
 - Okta
 - OneDrive for Business
 - Outlook Online
 - Salesforce
 - Salesforce Classic
 - Salesforce Lightning
 - ServiceNow
 - SharePoint Online
 - Webex
 - Zoom

4.3.2. Defining Custom Applications and Probes for ZDX

Custom applications allow you to apply ZDX monitoring to applications that you run in your private data center or that are hosted in the cloud. These can be custom-built applications, privately hosted instances of applications, or cloud services.

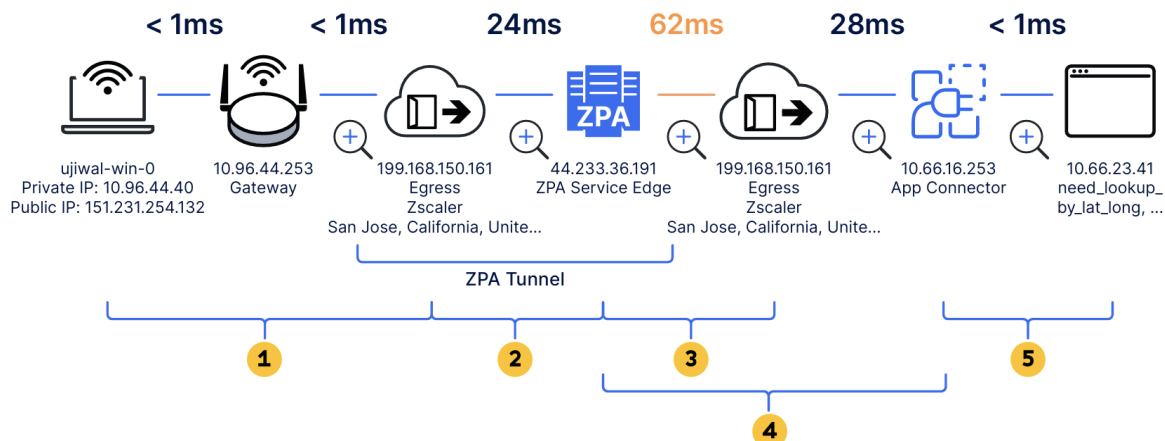
When you add a custom application, you must also add the associated probes to monitor the application. There are two types of probes you will build: a web probe and a Cloud Path probe. For each application, you will configure each probe type.

It is recommended that you build your web probes first, as the Cloud Path probe can follow the web probe and simplify configuration. Each probe has a frequency timer determining how often the probe should run.

4.4. ZDX, ZPA and Private Applications

ZPA offers the fastest, most secure access to private applications, services, and operational technology (OT) devices with the industry's only next-gen Zero Trust Network Access (ZTNA) platform.

ZPA allows you to make applications and OT devices invisible on the internet, reachable only through the ZPA service for authenticated and authorized users. When working with ZDX, ZPA allows you to leverage your existing authentication configuration and administrative accounts



When you examine a user's path using ZDX with ZPA, there are 5 groupings of segments to consider as you begin your analysis:

1. **User Device to Egress Point** – This segment covers your user's device and the network gateway, up to the service provider edge.
2. **Egress Point to ZPA Public Service Edge** – This segment is the ISP network between your end user and the ZPA Public Service Edge.
3. **ZPA Public Service Edge to Application Egress Point** – This is the segment between the ZPA Public Service Edge and the egress point from the application's point of view.
4. **ZPA App Connector to ZPA Public Service Edge** – This segment contains segment 3, as well as the internal network between the ZPA App Connector and the ZPA Public Service Edge.
5. **ZPA App Connector to Application** – This segment is the internal network between the ZPA App Connector and the private application the user is trying to reach.

4.5. Device Scoring and Device Events

Without visibility into device events, IT support teams are forced into a reactive mode—waiting for users to report problems, which leads to increased downtime and user frustration. Device scoring evaluates CPU, memory, disk, Wi-Fi, battery, and critical events like crashes.

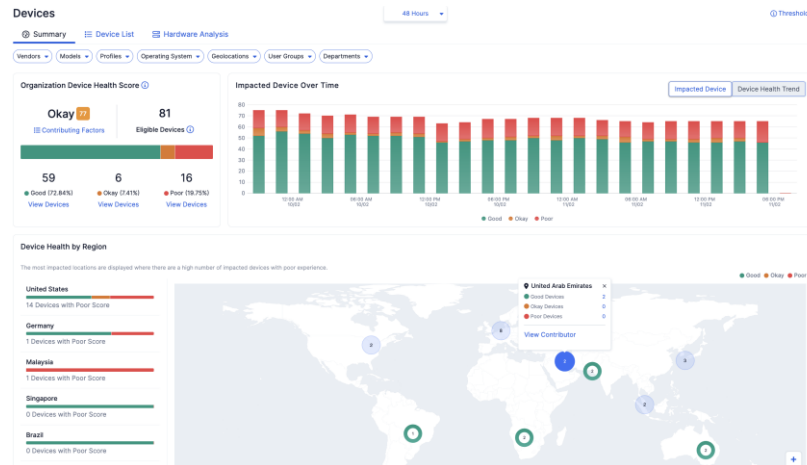
This approach not only delays issue resolution but also overwhelms service desks with preventable incidents. Adding to the challenge, critical device data is often spread across multiple external tools, creating silos and inefficiencies that slow down response times.

By implementing centralized device scoring and event tracking, organizations gain real-time insights into device health, enabling proactive support, faster root-cause analysis, and improved overall IT efficiency.

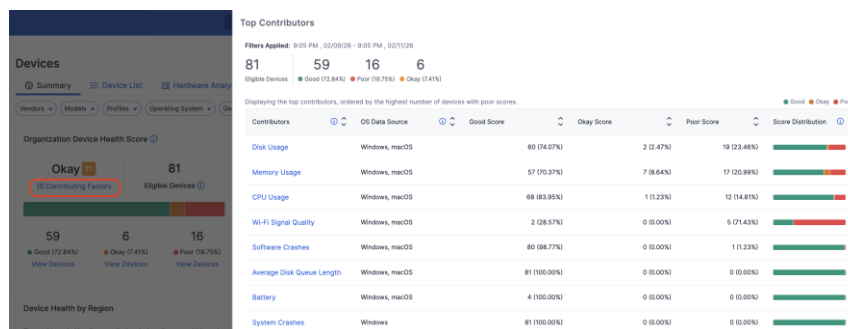
The Device Health dashboard uses existing user and device data to create a comprehensive device health scoreboard that monitors the performance of Windows and macOS devices. The dashboard provides useful data in evaluating trends to identify and understand the root causes of poor-performing devices. Then you can proactively collect device data and identify which devices require an IT's attention to remedy the device's poor performance (e.g., high CPU usage). With a dedicated dashboard, you can target hardware upgrades based on the device's usage for greater cost-effectiveness.

The ZDX's Device Health Score provides a comprehensive view of struggling devices across an entire organization, department, user group, or location. The score is determined by individual contributing metrics (e.g., CPU usage, memory usage, disk usage). The Device Health Scores are categorized as:

- **Good:** The usage profile has an acceptable Device Health Score and ranges from 66-100. The color for this range is green.
- **Okay:** The usage profile has an acceptable Device Health Score and ranges from 34-65. The color for this range is amber.
- **Poor:** The usage profile has a below an acceptable Device Health Score and ranges from 0-33. The color for this range is red.

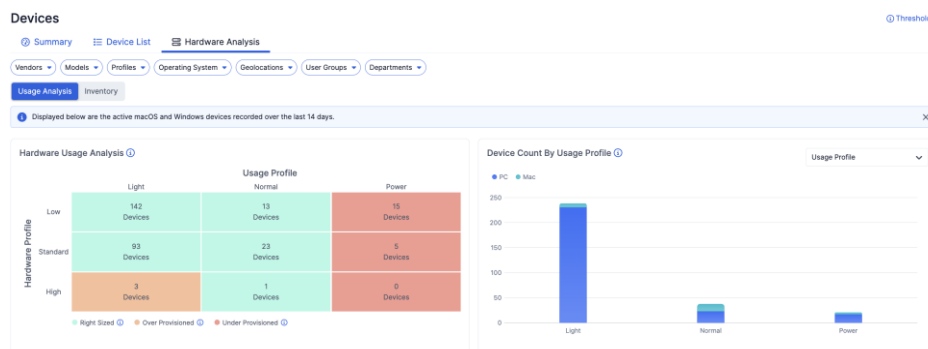


On the Summary page clicking on the “Contributing Factors” will view of what’s impacting device performance the most. It highlights key issues like high CPU usage, frequent crashes, or poor Wi-Fi signal in descending order of Poor score weightage.



The list of impacted devices can also be viewed by using the “Device List” option.

ZDX also offers a Hardware Usage Analysis displaying hardware profile based on hardware allocation and usage profile based on users and device usage trend.



5. Bill of Quantity

#	Product Name	SKU	Quantity
1	ZIA Business Edition	ZIA-BIZ-EDITION	2,719
2	Extended Log Store	ZSC-LOG-EXTSUBS	2,719
3	ZDX Advanced Plus	ZDX-ADV-PLUS	2,719
4	Inline Cyber Threat Protection — Sandbox Advanced	ZS-CTP-SANDBOX	2,719
5	Inline Cyber Threat Protection — Firewall Advanced	ZS-CTP-FIREWALL	2,719
6	Secure AI Access	ZS-AI-ACCESS-PRE	2,719
7	Zero Trust Browser Advanced	ZT-BROWSER-ADV	29,022
8	Zscaler Private Access Platform	ZS-ZPA-PLATFORM	200
9	ZPA for Legacy Apps (VoIP, S2C, etc.)	ZS-ZPA-VPN	200
10	Zscaler Private Access: Segmentation	ZS-ZPA-SEGM	200
11	Zscaler Test Environment	ZS-TEST-ENV	1
12	Zscaler Support Plus	ZCES-SUP-PLUS	1

The following workstreams are in scope for this engagement:

#	Workstream	Description
1	Identity Integration & ZCC Enrolment	Configure Zidentity federation with the customer's IdP (SAML 2.0 / SCIM). Enrol all 2,719 users via Zscaler Client Connector (ZCC) with automated device enrolment and user provisioning.
2	ZIA Business Edition Baseline	Tenant baseline, location hierarchy, SSL/TLS inspection policy, URL filtering, DNS security, Cloud Firewall standard ruleset, and Cloud App Control visibility, backed by Extended Log Store retention.
3	Cyber Threat Protection — Sandbox & Firewall Advanced	Configure Advanced Sandbox for unknown file analysis and zero-day protection, and Advanced Cloud Firewall policies for inline threat prevention across all users.
4	Secure AI Access	Discover and classify Generative-AI application usage; configure granular access, isolation, and DLP controls for sanctioned and unsanctioned AI tools.
5	Zero Trust Browser Advanced	Deploy browser isolation policies for high-risk categories, unmanaged devices, and third-party/BYOD access, covering the licensed browser session pool.
6	ZDX Advanced Plus	Deploy Digital Experience Monitoring across 2,719 users — endpoint, network, and application performance monitoring with proactive alerting.
7	Zscaler Private Access (ZPA)	Deploy ZPA Platform, Segmentation, and Legacy App support (VoIP/S2C) for 200 users — private application access, micro-segmentation, and legacy protocol enablement.

5.1. Identity Integration & ZCC Enrolment

- Configure Zidentity federation with the customer's identity provider (Azure AD, ADFS, or equivalent) via SAML

- 2.0 / SCIM.
- Provision user and group objects via SCIM for automated user lifecycle management.
- Deploy Zscaler Client Connector (ZCC) to all 2,719 users — enrolment via MDM (Intune/SCCM) or manual deployment per the customer's endpoint management capability.
- Validate ZCC enrolment, tunnel-up status, and ZIA policy enforcement.

5.2. ZIA Business Edition Baseline

- Tenant baseline: organisation hierarchy, locations, and sub-locations aligned to network topology.
- SSL/TLS inspection: decryption policy, bypass list, and certificate deployment guidance.
- URL filtering: category-based policies aligned to acceptable use requirements.
- DNS security: DNS control, sinkhole policy, and DNS-layer threat prevention.
- Cloud Firewall: standard outbound traffic controls and application-aware policies.
- Cloud App Control / Shadow IT visibility: application discovery and risk-tiered reporting.
- Extended Log Store activation for extended retention and reporting.

5.3. Cyber Threat Protection — Sandbox & Firewall Advanced

- Configure Advanced Sandbox policies for unknown file analysis, malware, and zero-day threat prevention.
- Configure Advanced Cloud Firewall policies for inline threat protection across all user traffic.
- Tune sandboxing quarantine and file-type inspection rules with the customer's security team.
- Validate threat protection policies with test scenarios before production enforcement.

5.4. Secure AI Access

- Discover and classify Generative-AI application usage across the user base.
- Configure granular access controls (allow / isolate / block) for sanctioned and unsanctioned AI tools.
- Configure DLP controls for sensitive data entered into AI prompts and file uploads.
- Produce an AI application risk visibility report for customer security review.

5.5. Zero Trust Browser Advanced

- Configure browser isolation policies for high-risk categories and unmanaged/BYOD access.
- Configure session controls: copy/paste, upload/download, watermarking, and clipboard restrictions as required.
- Validate isolation policies against the licensed browser session pool.

5.6. ZDX Advanced Plus

- Deploy ZDX across all 2,719 users for endpoint, network path, and application performance monitoring.
- Configure ZDX Score baselines, alerting thresholds, and Deep Tracing for priority applications.
- Provide initial digital experience baseline report to the customer IT team.

5.7. Zscaler Private Access (ZPA)

- Deploy ZPA Platform: App Connectors, App Segments, Access Policies, and Client Connector integration for 200 users.
- Configure ZPA Segmentation for micro-segmentation of private application access.
- Configure ZPA support for Legacy Apps (VoIP, Server-to-Client, and other non-web protocols) as required.
- Validate private application access, connector high availability, and policy enforcement.

6. Assumptions

- The Zscaler tenant is provisioned by Zscaler and Admin Portal access is granted to Starlink PS engineers before Week 1 kick-off.
- Customer holds valid Zscaler licenses (as per the BOM in Section 2) for all in-scope users before the engagement commences.
- IdP details (SAML metadata, SCIM endpoint, AD group structure) are provided before Month 1 configuration begins.
- The customer's endpoint management team provides MDM/SCCM access or deployment support for ZCC rollout.
- Customer IT is responsible for deploying ZCC to endpoints under Starlink's guidance; remote desktop access or screen-sharing support is made available for troubleshooting.
- All remote sessions are conducted over a stable video/screen-share platform agreed between parties (Teams/ Zoom/WebEx).

7. Dependencies

- Timely provisioning of the Zscaler tenant, ZIA/ZPA/ZDX admin console access, and license activation by Zscaler prior to kick-off.
- Customer's network team to confirm and, where required, update firewall/proxy egress rules to support ZIA, ZPA App Connector, and ZDX cloud connectivity.
- Customer's IT security team to define DLP, AI-access, and browser-isolation policy requirements within the timelines set out in Section 5.
- Customer's endpoint management team (Intune/SCCM/MDM) to have management coverage sufficient for ZCC deployment to the in-scope device population.
- Availability of Customer IT and business stakeholders for design workshops, pilot testing, and UAT sign-off within the agreed project plan windows.
- Starlink to coordinate commercial, licensing, and escalation matters between Starlink and Customer.
- Any delay in the above dependencies may impact the 3-month timeline and is subject to a Change Order.

8. Out of Scope

- Zscaler modules or SKUs not listed in the Bill of Materials in Section 2.
- On-premises hardware installation, network re-architecture, or firewall/router changes beyond the egress rule guidance required for Zscaler connectivity.
- Active Directory / Azure AD tenant configuration — the customer owns IdP administration; Starlink provides integration guidance only.
- End-user helpdesk, device support, or ZCC troubleshooting beyond initial deployment validation.
- Custom application development or non-standard API integrations beyond the packaged Secure AI Access and CASB-style connectors covered in Section 3.
- Additional users beyond 2,719 (ZIA/ZDX/CTP/AI Access/ZTB) or 200 (ZPA) without a Change Order.
- Any effort beyond the 3-month engagement duration without a signed Change Order.
- Procurement or licensing of Zscaler or any other product.

On-site presence — this is a fully remote delivery engagement

9. Place(s) of Performance

CNS-StarLink will implement the above solution. CNS-StarLink will update the customer once per week (day and time TBD) for a weekly status meeting based on requirement.

CNS-StarLink Resource(s) work a normal workday of (8) hours and will adhere to Customer's local business hours. In addition, CNS-StarLink resources will adhere to the local CNS-StarLink office holiday schedule. Any services performed after hours (8 pm to 8 am) and on weekends must be approved in advance by CNS-StarLink Professional Service Management and will be assessed 1.5 x per hour worked. For example, if a resource is requested to work after hours for an additional two (2) hours, three (3) hours will be charged against any existing PS Consulting days or if additional time is needed, a project change request and purchase order will be required to cover the additional time required before exceeding the PS Consulting days purchased.

If Customer reschedules the services within two (2) weeks of the scheduled services, customer will be charged two (2) full business day from any existing PS Consulting days. If additional time is required to cover the cancellation or rescheduling of the services a project change request and purchase order will be required to cover the additional costs.

10. Deliverables

The following deliverables will be produced during the project's timeframe:

- Implementation & Deployment of solution components in production.
- High level Design Documentation
- As-Built documentation.

The Project Engineer will share daily updates of the progress to the end user and in case the end user does not provide confirmation within 2 business days then the delivery will be deemed as accepted by the end user.

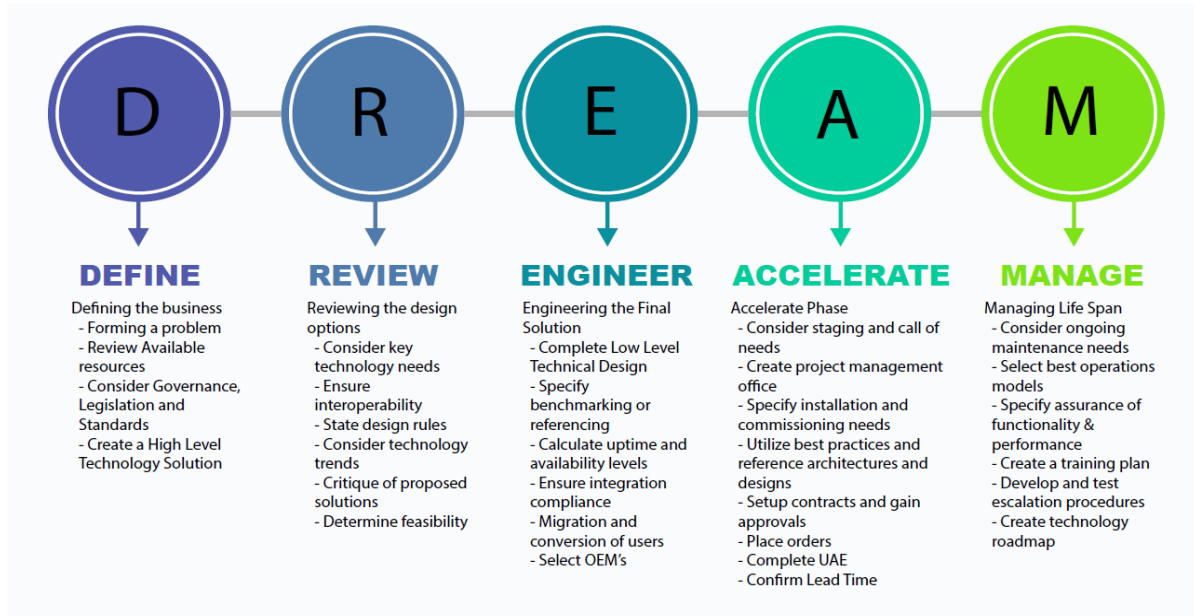
11. Miscellaneous

Project kick off meeting will be scheduled four (4) weeks from the date of the PO, provided CNS-Starlink has received signed SOW. Please note signed SOW needs to be shared with CNS-StarLink resource at least 2 weeks prior to the kickoff.

The PS Consulting days must be utilized within six (6) months from the last signature date set forth below ("SOW Effective Date"). Any hours/days remaining following the six-month period will expire, along with any obligation to deliver any further services unless otherwise mutually agreed to in writing.

12. Solution Implementation

12.1. Implementation - DREAM Methodology



DREAM is a proven service delivery methodology designed by CNS to ensure the success of all our technology and services engagements.

Through this tried and tested approach, we are able to provide significant benefits by working with our customers at a level that starts even before the consideration of ICT as part of the solution.

By engaging business and technical experts, we can help introduce not just independent experience and knowledge, but a CNS-developed step-by-step structured methodology that will increase the scope of the engagement and increase the quality and longevity of the final solution.

By using our structured DREAM methodology (**Define, Review, Engineer, Accelerate, Manage**) we have created an organization that can simplify end to end your ICT needs, big or small, starting from the design and staying with you throughout the your technologies lifespan.

Define, Review, Engineer, Accelerate, Manage are the five stages that CNS follows as a methodology to ensure that your ICT deployment meets in full the needs it sets out to satisfy. These stages work equally well for engagements of any size or type. The most effective utilization of the methodology is to implement it from the Define stage and to proceed step by step towards the final Manage stage.

12.2. Implementation & Migration Approach

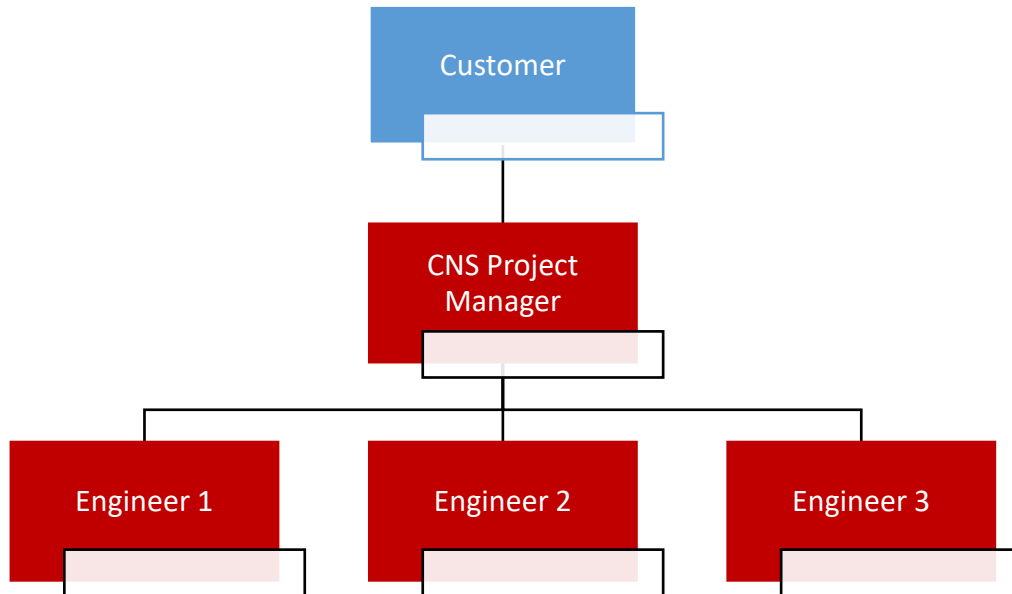
CNS-Starlink follows a phased implementation and migration approach designed to move the customer from current-state connectivity to a fully validated Zscaler Zero Trust architecture with minimal business disruption:

- Discover & Design — Gather network topology, IdP details, application inventory, and produce a Detailed Design Document (DDD) that is reviewed and signed off before configuration begins.
- Foundation Build — Stand up ZIdentity federation, ZIA Business Edition baseline, and begin phased ZCC enrolment in monitor-only mode to validate traffic visibility before enforcement.
- Security Layer Rollout — Progressively enable Cyber Threat Protection, Secure AI Access, and Zero Trust Browser Advanced policies, moving from alert-only to enforcement mode after validation with a pilot user group.
- Experience & Access Layer — Deploy ZDX monitoring in parallel, followed by ZPA application onboarding in controlled batches (pilot app segments first, then phased rollout to the full 200-user population), running alongside existing access methods until cutover is confirmed.
- Cutover & Decommission — Once pilot validation is complete for each workstream, extend enforcement to the full user population and coordinate decommissioning of legacy access paths (e.g., VPN) with the customer's change control process.
- Validate & Transition — Execute UAT across all workstreams, tune policies based on findings, and complete Knowledge Transfer and formal sign-off.

This phased, pilot-first approach is applied consistently across ZIA, ZPA, CTP, Secure AI Access, and ZTB workstreams to de-risk enforcement changes and ensure business continuity throughout the 3-month engagement.

13. Project Management

13.1. Project Structure



13.2. Proposed Team Structure

Role	Responsibilities	Mode
PS Lead Consultant	Owns all technical configuration across ZIA, ZPA, ZDX, CTP, Secure AI Access, and ZTB. Single point of technical accountability for the engagement.	Remote
Starlink Partner Coordinator	Facilitates Customer stakeholder access, IdP and network information gathering, and deployment logistics coordination.	Remote / Onsite
Customer IT Point of Contact	Provides access credentials, network information, endpoint management access, application inventory, and UAT sign-off.	Customer-side

Working Hours: Remote sessions are conducted during Customer standard working hours (Monday to Friday, 0900–1700 hrs). Out-of-hours sessions require advance written agreement from both parties.

13.3. Project Plan — 3 Months

All activities are delivered remotely by CNS-Starlink PS engineer.

Phase	Timeline	Key Activities	Milestone
1	Month 1 — Weeks 1–2	Kick-off with Customer IT and Starlink. Tenant provisioning verification. Collect IdP details (Azure AD/ADFS/Okta), network topology, user group structure, and application inventory. Confirm ZCC deployment method (MDM/SCCM/manual). Produce Detailed Design Document (DDD). Begin ZIdentity/SAML/SCIM federation and ZIA Business Edition baseline configuration.	DDD signed off · IdP details confirmed · ZCC deployment plan agreed
2	Month 1 — Weeks 3–4	Complete ZIA Business Edition baseline: location hierarchy, SSL/TLS inspection, URL filtering, DNS security, Cloud Firewall, Cloud App Control. Extended Log Store activation. ZCC enrolment rollout across 2,719 users. PAC file generation and validation.	ZIA Baseline live · ZIdentity active · ZCC enrolment underway
3	Month 2 — Weeks 5–6	Configure Cyber Threat Protection: Sandbox Advanced (file analysis, zero-day protection) and Firewall Advanced (inline threat prevention). Configure Secure AI Access policies for GenAI application visibility and control. Internal testing and validation.	CTP Sandbox & Firewall configured · Secure AI Access policies active
4	Month 2 — Weeks 7–8	Deploy Zero Trust Browser Advanced isolation policies. Deploy ZDX Advanced Plus across all 2,719 users — endpoint/network/application monitoring configured and validated.	ZTB Advanced deployed · ZDX Advanced Plus live
5	Month 3 — Weeks 9–10	Deploy ZPA Platform for 200 users: application segments, connectors, and access policies. Configure ZPA Segmentation and Legacy App support (VoIP/ S2C). Activate Support Plus coverage.	ZPA Platform, Segmentation & Legacy App access live for 200 users
6	Month 3 — Week 11	UAT across all workstreams with representative users. Collect and consolidate UAT findings. Policy tuning and false-positive remediation across ZIA, CTP, AI Access, ZTB, ZDX, and ZPA.	UAT complete
7	Month 3 — Week 12	Final validation across all workstreams. Knowledge Transfer (KT) sessions with customer IT admins. Deliver Operational Runbook, As-Built Configuration Document, and SOPs. Formal project closure review and sign-off.	All users live · KT complete · As-Built delivered · Project signed off

Note: Week-by-week timing is indicative across the 3-month engagement. Activities within a phase may run in parallel where dependencies allow. Final schedule is confirmed at Week 1 kick-off. Timeline extensions due to delayed information provision by the Customer are subject to a Change Order.

13.4. CNS PMO History

CNS Project Management Office was officially formulated in 2007 based on CNS commitment to Customer Service. Its main objectives are:

- Coordinates resources
- Apply best Practices and standards as per Project management institute.”
- Plan and execute Projects with high quality levels”

13.5. PMO Qualifications

CNS PMO consists of teams of experienced and certified Project managers.



CNS PMO has Project Management Professional (PMP) Certified Project Managers, who will ensure best practices are followed throughout implementation.

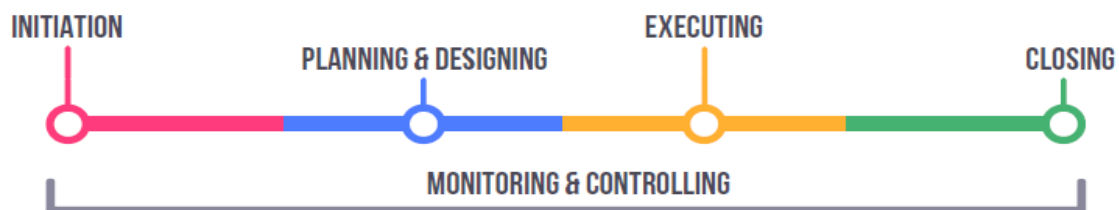
Project Manager	Profile
Projects Team Lead	• PMP Certified • +10 Years of IT experience • +5 years of Project Management experience
Project Managers	• PMP Certified • +10 Years of IT experience • +4 years of Project Management experience

13.6. PMO Methodology

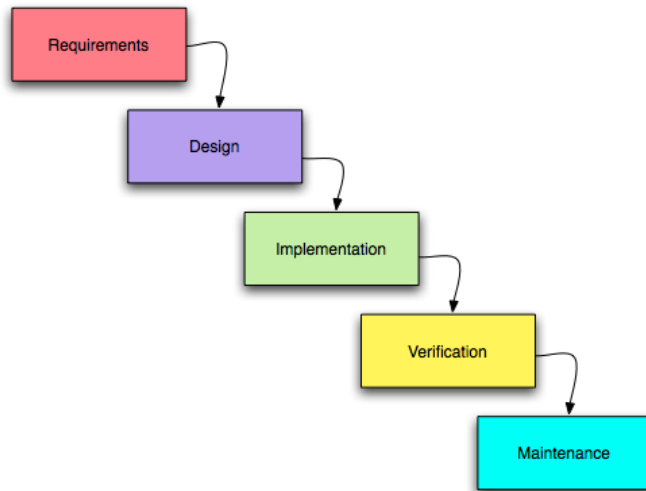
CNS developed an overall methodology to ensure effective management of complex projects, allowing for strong governance, co-ordination and control of the overall activities, resolve escalations and ensure appropriate resources are available to undertake the work.

CNS' Project Management Methodology is aligned with the principles set out by Worldwide Project Management Institute (PMI) as defined within "Project Management Body of Knowledge" (PMBOK).

PMBOK refers to the five process steps of project management, these are initiating, planning, executing, controlling, and closing.

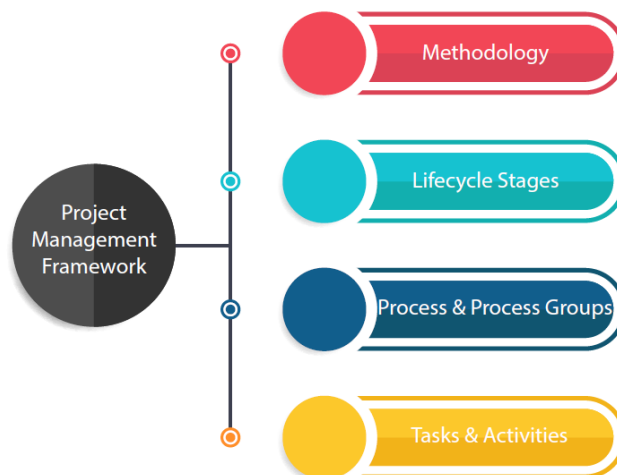


CNS applies the **Waterfall Project Management Methodology**, it is a project management methodology theme with a very simple approach that values solid planning, doing it once and doing it right. It's simple to understand because you simply make a good plan, and execute on it.

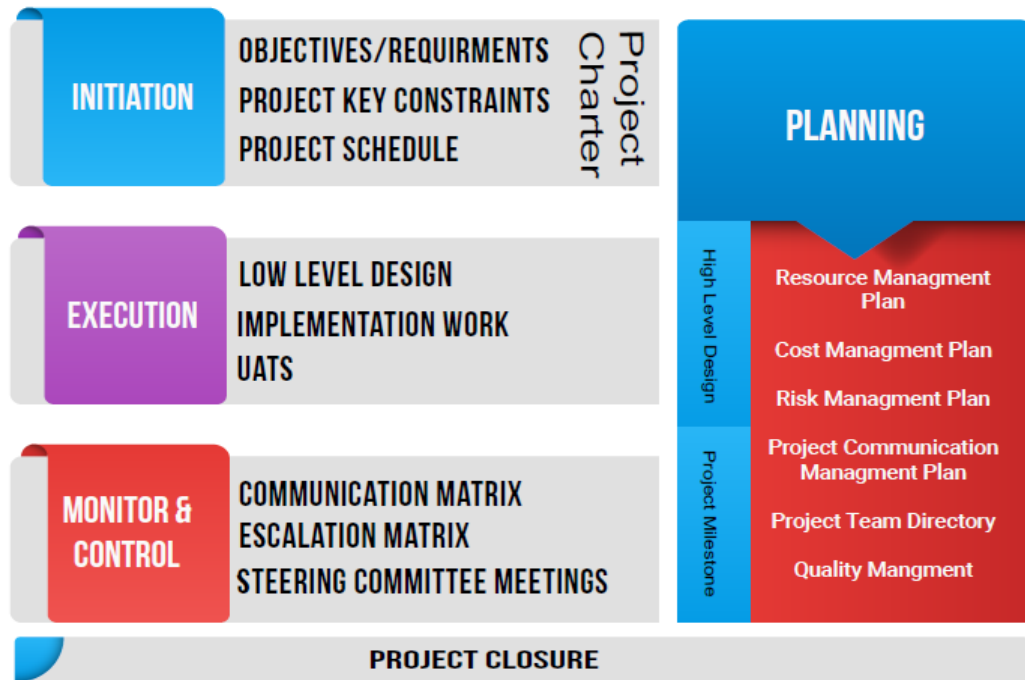


At the top of the waterfall, requirements are defined in full. Work then cascades like a waterfall in each phases of the project. Each phase must be completed before the next phase can begin without overlapping between phases. The outcome of one phase becomes the input for the next.

After the plan is approved, there's little scope to adapt the plan unless absolutely necessary, and changes that are needed usually require change requests. The project then flows through the process from requirements, through design, implementation, testing and into maintenance.



13.7. PM Process



13.7.1. Project Start-up

During Project Start-up the following activities will be performed

Review Bid and Contract

CNS will review Bid and Contract material, Identification of Stakeholders along with Risk assessment and documentation.

Review Project Assets with Customer

Review of Contract with customer, Review / Sign off of Project Management Plan (PMP) and update Project Change Management Strategy.

Develop Work plan

Develop baseline project plan (work plan) based on project scope including iteration plan, roles & responsibilities and dependencies.

Resource Plan

Resource management involves planning for, obtaining and managing the resources (including subcontractors, if applicable) required to perform the project successfully from the schedule, budget and quality points of view.

CNS will perform this internal activity for resource plan schedule and communicate the high-level dependencies required for resource plan or mobilization.

CNS mobilization period 3 to 4 weeks.

Establish Governance

CNS has a strong emphasis on Project governance and this phase will establish the governance structure including Scope Change Management Process, Financial Management, Risk/Issue management, Resource

Management, Communication Plan, Quality management, Documentation and Reporting process, Configuration Management, Infrastructure Management. The PMP will define the governance structure.

Projects and programs are managed according to governance and oversight policies and processes as we strive to effectively manage and mitigate risk to our projects and programs. Ensuring appropriate visibility and governance to our projects to achieve the greatest business value for both CNS and for our Customer.

CNS management will have representation in the project steering committee and present the progress on monthly basis along with identified or potential risks/issues. The escalation process shall also be defined and agreed upon as part of Project Management Plan (PMP) during the startup phase.

13.7.2. Project Execution and Control

Project Execution and Control is an activity during the full life cycle of the project across all phases. This activity will ensure the project is on schedule, on budget and on agreed scope. Any impact will be discussed with customer on case to case basis. In addition, progress reports from CNS will provide customer the Change Control Board (CCB) information required to track the status of the project.

A steering committee by both CNS and customer is required. Frequency of this will be decided mutually during project kick-off.

Manage Scope, Acceptance and Approvals

CNS will track any changes to the scope of work including deliverables, schedules, and resources. In the event of a change, a change request will be raised, discussed with customer and confirmed by both customer and CNS. Change request process include impact assessment, discussion between CNS and customer manager, and approval by Steering committee and possible revision of Project plan.

There is a need for CNS to obtain documented acknowledgement that a client has received a deliverable and has either accepted it or rejected it. The document that is used for this is a Certificate of Acceptance or a Certificate of Acceptance/Rejection. The benefit of this latter form is that it requires the client to specify any reason(s) for having withheld acceptance. Acceptance criteria for all phases and deliverables will be agreed upon by customer and Oracle during the initiation phase.

It is possible that during the project its scope may be changed or the schedule may be amended. In addition, customer may request additional assistance from CNS for tasks that are not under the agreed scope. In order to control and manage such changes, any request for additional CNS work will be formally submitted to the CNS Project Manager using a Change Request form. CNS will evaluate the request and will agree the commercial and schedule implications with customer. Once agreement is reached and the Change Order is signed, the additional work will be incorporated into the Project Plan and CNS schedule, estimates and fees will be adjusted accordingly.

Any large or complex changes may incur charges for impact analysis effort. CNS will agree a budget with customer for an impact analysis prior to investigating the requirements in detail.

Manage Project plan / Work plan

An estimated project plan identifying activities and schedule is provided by CNS as part of this proposal. The final project plan will be part of the deliverable from CNS. CNS will review and facilitate sign off from customer Program Manager with contractual ownership of overall project plan with CNS.

The final signed off version will be used as baseline version and used for all further monitoring and controlling activities. Any changes required on the project plan will need to be agreed in writing by CNS and customer.

The overall project plan is maintained by CNS but individual activities within a phase may be dependent on customer or 3rd party vendor. CNS will provide inputs to the project plan including activities related to customer.

Customer activities which are independent of CNS work need to be maintained separately by their project manager. CNS will discuss with customer Program Manager on the expected completion schedule of these activities and possible impact on other activities and overall project duration if there are delays.

Risk Management

Effective risk hedging and risk management is a key factor for successful projects; to this aim a full set of risk evaluation tools have been developed to systematically identify, evaluate and mitigate overall project risks.

Risks and issues that are identified throughout the project will be recorded in **Risk and Issue Registers**. This includes not only software or enhancement issues but also scheduling issues, policy/procedure items, and any other activity or decision that may affect completion of the project. Customer and CNS will review the risks and issues as part of the regular Status and Project Board meetings. Customer and CNS will work together to mitigate the risks and resolve the issues.

Manage Communication

Throughout the project it is necessary to understand who the key stake holders are, both within the client organization and within CNS, and to communicate to them information each individual should receive. This information would normally be the project status, plans, risks and other key attributes. Issues that are to be escalated will flow through the hierarchy of project organization / structure. In addition, senior management of CNS Utilities practice will be available at all times for escalations, not limited to the project but all the product.

Communication plan as part of PMP will include the Status Reporting procedures, communication hierarchy and channels between CNS, Customer and other third party vendors if applicable.

Quality Management

During Planning Phase, CNS will identify any performance or quality standards that must be met upon approval of the final results of the project and develop acceptance criteria for the final work product.

Upon completion of a deliverable, a quality review will be conducted to evaluate the deliverable, and compare the results against acceptance criteria. Upon Project completion a final quality check is performed to verify that the end product meets the quality standards set in UAT Plan

13.7.3. Project Closure

Gain Acceptance

Project Acceptance Report will provide an acceptance on overall project and summarizes the progress. All prior acceptance certificates throughout the lifecycle of the project is attached in the report to form the complete acceptance of project.

Close Processes and Contract

This is formal process to close the contract, release resources, submit final reports, close financials including invoicing and payments, establish a follow-up process if required and submit recommendation of future enhancements.

14. Service Delivery (Support and Maintenance)

14.1. What CNS Support Services are:

- Hardware Support
 - Technical problem investigation
 - Remote diagnostics
 - Unlimited on-site hardware parts replacement
 - Firmware updates (push & pull)
 - Preventive maintenance
 - Access to knowledge data bases
- Software Support
 - Technical problem investigation
 - Remote diagnostics
 - Unlimited on-site hardware parts replacement
 - Firmware updates (push & pull)
 - Preventive maintenance
 - Access to knowledge data bases

14.2. CNS Customer Happiness Center



- 365 x 7 x 24 Operations
- Single Point of Contact for Customer Related Requirements
- Reachable per Email, Phone & CNS Portal
- Equipped with Best in Class Technologies
- Highly Available & Complete DR
- Call Logging/Ticketing ICT & EFT
- NOC / ICT Monitoring
- 1st & 2nd level Support End Users
- 1st level Support for ICT Infrastructure
- NOC / ATM Monitoring
- 1st level Support ATM

14.3. CNS Resources

CNS Provides you with the best skills through our certified engineer to ensure your projects are delivered at the highest quality and within the timeline

# of Capable Resources	Product	Vendor	Comment
8	Network & Communications	Cisco	Certified Partner
8	Servers	HPE, Oracle, Cisco	Certified Partner
8	Storage	HPE, Oracle, EMC	Certified Partner
5	Backup Solutions	Veeam, HPE, CommVault, Symantec	Certified Partner
4	Cyber Security / Firewalls	Cisco, Palo Alto, FortiNet	Certified Partner
10	Virtualization Solutions	Microsoft, VMWare, Oracle	Certified Partner
3	Load-Balancer	F5	Certified Partner
4	End User Security	Symantec, FortiNet	Certified Partner
12	ITSM / NOC Tools	Assyst, Manage Engine	Certified Partner
4	Hybrid Cloud	Microsoft, VMWare, Oracle	Partially Certified

14.4. Escalation Matrix

Level	S. No	Name	Designation	Contact No.	Email ID
LEVEL 1	1	IT Service Desk	Single Point of Contact	Direct: +971 4 2384455	ITSD@cns-me.com
LEVEL 2	1	Fadi Nasser	Senior Account Manager	Mobile: +971 55 807 6376	Fadi.nasser@cns-me.com
	1	Nauman Iqbal	Technical Team Lead	Ext # 8522 Mobile: +971558498783	Nauman.Iqbal@cns-me.com
LEVEL 3	1	Ravi Srivastava	Director - MS & ITO	Mobile: +971 55 158 4043	Ravi.Srivastava@cns-me.com

15. Appendices

15.1. Proposal Terms and Conditions

Proposal Validity : 30 Days

Delivery and Implementation

Delivery Date : As per above Place of Performance

(That Delivery Deadlines provided is as per the current delivery dates received from OEM/ Supplier which are subject to further changes/ revisions from time to time as received from the OEM/ Supplier. However, CNS is fully committed to keep the project representatives informed of these delays and their effect on overall job completion. CNS will diligently seek to minimize the delays to the best of its ability and the effects of these delays on its Customer and the Project)

Implementation : As per above SOW

15.2. CNS Company Profile

We are a UAE based system integrator and managed services provider that offers business solutions and services to clients of telecommunication, governments, financial services, education, energy and enterprises across the GCC.

Founded in 1987 CNS employs has **offices across the Middle East**, we have an extensive reach into, and **regional knowledge of the market cultures** of the Middle East.

A member of **the Ghobash Group**:

- Provides an access infrastructure for the best business opportunities across the region.
- Personally committed to **developing new partnerships and investments** that support economic and social prosperity.
- Embraces a corporate culture that fosters excellence, team spirit, and **creative thoughts and solutions that help drive our growth**.

CNS Contact Information	Description
Company name	Computer Network Systems - CNS
Established	1987
Contact Person	Fadi Nasser
Holding Company	ABAN Investment Company
Address	DXB Office: Makeen Bldg, Airport Rd, P.O Box 52137, Dubai, UAE AUH Office: Madinat Zayed Office Tower, 12th Floor, Sultan Bin Zayed the First Street, P.O. Box 46144, Abu Dhabi, UAE Oman Office: Pearl Tower, Ground floor Ghala, Muscat, Sultanate of Oman. PO Box: 1718, PC 130 Azaiba Kuwait Office: 2nd Floor, Dar Al Awadi Centre P.O. Box 29927, Sharq, Kuwait KSA Office: Riyadh-Al Sulimaniah- Prince Mamdoh Bin Abdul Aziz street-Al Safwah Trading Center office no. 72
Telephone number	DXB: +971 04 596 4333 AUH: +971 (0) 2 508 7888 Oman: +968 2411 1500 Kuwait: +965 2232 2999 KSA: +966 558682818
Fax number	DXB: +971 4 238 4848 AUH: +971 2 644 0151 Oman: +968 2411 1501 Kuwait: +965 2232 2990 KSA: +966 114734521
Email address	info@cns-me.com
Total Years of Experience for the Company	35+ Years
Main Activities and business area	• Network and Communications • Data Center and Hyper converged Infrastructure • Applications and Software • Cybersecurity • Cloud • Banking • Digital Transformation • Emerging Technologies • Managed Services and Engineering services
No. of employees (Full time)	800+
Offices	• Dubai, United Arab Emirates • Abu Dhabi, United Arab Emirates • Muscat, Sultanate of Oman • Kuwait • Riyadh, KSA • India (Offshore) • Egypt (Offshore)

15.2.1. Ghobash Group

The Ghobash Group stands among the most diversified, privately-owned local conglomerates in the United Arab Emirates with multiple investments and operating companies in seven major industries.

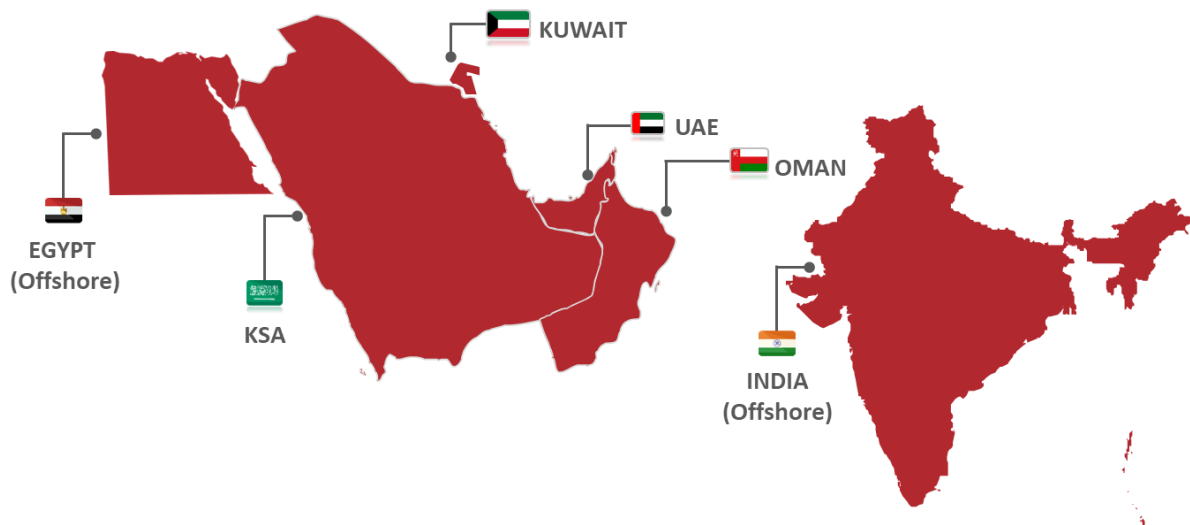
The Ghobash Group extends over 50 collective years of regional insight to investors seeking to tap into the lucrative market opportunities of the rapidly developing MENA region.

The Ghobash Group has spent over 40 years building a name now among the most valued in the market. A name earned by creating sustainable business-equity according to the values, ethics, and insights of the UAE and surrounding region.

With an auspicious past, Ghobash Group embraces a progressive outlook to the future, offering like-minded partners and stakeholders our key to the Middle East and beyond.

Diversity	Experience	Territory
19 operating companies in 7 major industries	50+ Years of collective local knowledge and insight	Strategic footprints in 5 GCC countries, East Africa & South East Asia

15.2.2. CNS Territory



15.2.3. CNS Vision and Mission

Vision

CNS empowers organizations to lead in the digital future.

Mission

We are committed to helping clients achieve mission-critical business objectives by supporting their digital transformation and innovative solutions with best-in-class services.

15.2.4. CNS Solution Key Differentiators

One-Stop-Shop - Innovative solutions and best in-class services across verticals



24/7 Customer Support



800 + Digital Professionals +
In-House Software Team

ISO 9001:2015 – Quality Management Systems
ISO/IEC 27001:2022 – Information Security Management Systems
ISO/IEC 20000-1:2018 – IT Service Management Systems
ISO 37001:2016 – Anti Bribery Management Systems

Customer can be assured of the below:

- Full Handover and Knowledge Transfer of each solution provided and the integration between solution components
- Documentation of the solution components and how each component interact with each other.
- From single consoles the client will be able to manage all the environment

15.2.5. CNS Value Proposition

We Have Proven Experience

We have extensive experience successfully delivering similar initiatives for other clients of similar size in both Public Sector and other industries. Specific examples of our delivery skills and their impact include:

- Skills and experiences - enables quick start up and higher quality results
- Credentials and experience with similar solutions - enables incorporation of latest thinking and best practices
- Experience at scale – lowers delivery risk

We will assume accountability for outcomes, service levels, and results.

CNS is committed to delivering real, measurable outcomes, in the expected timeframe, at the agreed-to service levels, for the agreed to price.

- In all aspects of our relationship, we will recognize and respond to your business goals
- We will report accountability for our jointly established goals to client on a regular basis.

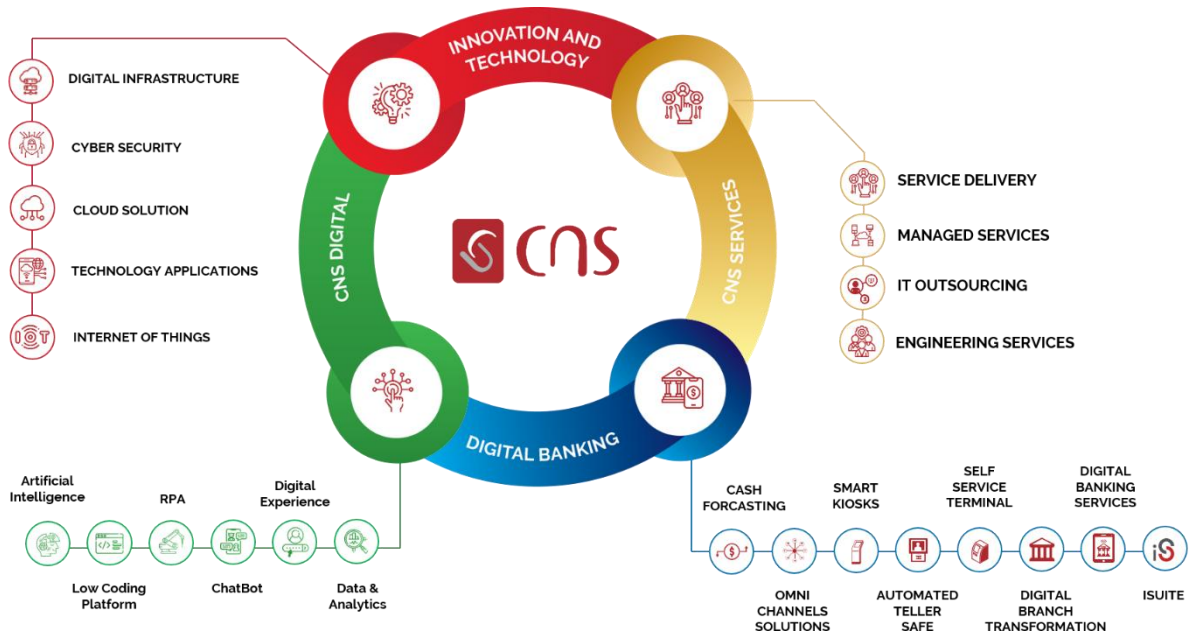
15.2.6. Our Recent Awards

 FERTINET Certificate of Appreciation for Winning a Strategic Project	 هيئة الصحة بدبي DUBAI HEALTH AUTHORITY Excellent Performance Certificate 2020	 CISCO Social Impact Partner of the Year 2021	 Diebold Nixdorf Innovative Project of the Year 2021	 NetApp Technology Partner of the Year 2021
 امارات Emarat Appreciation Award 2021	 Best System Integrator of the Year 2021	 KnowBe4 Human error. Conquered. Partner to Watch 2021 Middle East	 دبي الاحمر Certificate of Gratitude and Recognition 2021	 Diebold Nixdorf Best Strategic Deal Award of the Year 2022
 Zero Trust Security Partner of the Year 2022	 PhishRod Emerging Partner of the Year 2022	 software.com Best New Partner of the Year 2023	 NEW AGE BANKING SUMMIT OMAN 31 MAY 2023 SHERATON OMAN HOTEL Digital Transformation Partner of the Year for BFSI Award 2023	 Best Cybersecurity Solutions Partner of the Year 2023
 وزارة الصحة ووقاية المجتمع MINISTRY OF HEALTH & PREVENTION Certificate of Gratitude and Recognition 2023	 Gulf IT Rook's Resilience Award 2023	 Highest Cybersecurity Potential of the year 2023	 بلدية دبي Dubai Municipality Certificate of Appreciation 2024	 NetApp Cyber Resilience Partner of the Year 2024

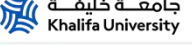
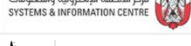
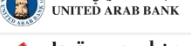
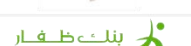
15.2.7. Our Portfolio

CNS has always been on the forefront of innovative technology solutions, bringing our customers closer to leading edge dynamic solutions.

CNS is focused on providing business agility, security and speed quality in Digital Transformation



15.2.8. Our Clients

 RTA مركز المعلومات والمعلومات SYSTEMS & INFORMATION CENTRE	 Dubai Healthcare City	 ADNOC	 Khalifa University	 SHARJAH ISLAMIC BANK
 SEHA مركز الخدمات الصحية HEALTH SERVICES CENTRE	 Ministry of Health	 enoc	 MOHAMMED BIN RASHID UNIVERSITY OF MEDICINE AND HEALTH SCIENCES	 UNITED ARAB BANK
 FEDERAL AUTHORITY FOR IDENTITY & CITIZENSHIP	 Children's Hospital	 Emarat	 Hamdan Bin Mohammed Smart University	 bank muscat
 Dubai Municipality	 AMERICAN HOSPITAL	 Sanad A Mubadala Company	 American University of Sharjah	 Sohar
 Emirates Health Services	 TDRA هيئة تنظيم الاتصالات والحكومة رقمية TELECOMMUNICATIONS AND DIGITAL GOVERNMENT REGULATORY AUTHORITY	 SCOPE	 ZAYED UNIVERSITY	 Bank Dhofar
 MINISTRY OF HEALTH & PREVENTION	 etisalat	 noon	 University of Niwa	 ADIB
 MINISTRY OF INTERIOR	 du	 International Securities an IFC company	 UAEU الجامعة العربية للعلوم United Arab Emirates University	 NBO
 DUBAI POLICE	 Dubai Electricity & Water Authority	 MAJID AL FUTTAIM	 NAKHEEL	 Al Maryah Community Bank
 ABU DHABI POLICE	 Ministry of Health	 BAT MIDDLE EAST	 EMICOOL	 CBI
 DEPARTMENT OF HEALTH	 Ministry of Health	 Federal Electricity & Water Authority	 EMICOOL	 alizz islamic bank

15.2.9. Partner and Alliances